



ATLAS CONF Note

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Measurements of the Higgs boson production, fiducial and differential cross sections in the 4ℓ decay channel at $\sqrt{s} = 13$ TeV with the ATLAS detector

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Inclusive, differential and production mode cross sections of the Higgs boson production in proton-proton collisions are measured in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel. The proton-proton collision data were produced at the Large Hadron Collider at a centre-of-mass energy of 13 TeV and recorded by the ATLAS detector from 2015 to 2017, corresponding to an integrated luminosity of 79.8 fb^{-1} . The inclusive fiducial cross section for the process $H \rightarrow ZZ^* \rightarrow 4\ell$ is measured to be $\sigma_{\text{fid}} = 4.04 \pm 0.47 \text{ fb}$, while the Standard Model prediction is $\sigma_{\text{fid,SM}} = 3.35 \pm 0.15 \text{ fb}$. The cross-section times $H \rightarrow ZZ^*$ branching ratio for gluon fusion and vector-boson fusion production are measured to be $1.22 \pm 0.18 \text{ pb}$ and $0.25 \pm 0.09 \text{ pb}$, respectively. All measurements are in agreement with the Standard Model prediction.

In the June 2 2018 version, a double counting of the observed event in the VH leptonic category and the use of a previous version of the hadronic jet calibration in some parts of the analysis were found. All of the affected results have been corrected. These changes only concern the production cross section and STXS results. Figures 12, 13, 14, 19, 20 and Table 9 have been updated.



1 Introduction

The observation of the Higgs boson by the ATLAS and CMS experiments [1, 2] with the LHC Run 1 data at centre-of-mass energies of $\sqrt{s} = 7$ TeV and 8 TeV has been a major step towards the understanding of the mechanism of electroweak (EW) symmetry breaking [3–5]. Further studies of the spin, parity, couplings, fiducial and differential cross sections of the new particle have shown no significant deviation from the predictions for the Standard Model (SM) Higgs boson with a mass near 125 GeV [6–18].

This note presents an updated measurement of the Higgs boson properties in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel (where $\ell = e$ or μ) using $79.8 \pm 1.6 \text{ fb}^{-1}$ of proton-proton (pp) collision data collected at $\sqrt{s} = 13$ TeV by the ATLAS detector between 2015 and 2017. The average event pile-up - *i.e.* the mean number of interactions per bunch crossing - is 32 for this dataset. Three types of measurements are presented in this note: measurements of inclusive and differential production cross sections in fiducial phase space and measurements of cross sections using the SM production modes as “templates” in simplified fiducial regions. All measurements are performed under the assumption that the Higgs boson mass is 125 GeV, and are compared to Standard Model predictions.

The main differences with respect to the previous publications [11, 12] are: (i) larger integrated luminosity, (ii) improved event reconstruction, (iii) improved electron reconstruction [19] and (iv) additional reconstructed event categories in the analysis to enhance the sensitivity to the production of the SM Higgs boson associated with a vector boson (VH , $V \rightarrow \ell\nu/\nu\nu$) and with a top quark pair ($t\bar{t}H$).

The note is organised as follows: a brief introduction of the ATLAS detector is given in Section 2. In Section 3, the data and simulated signal and background samples are described. The selection and categorisation of the Higgs boson candidate events is detailed in Section 4, while the background modelling is described in Section 5. The analysis strategy is outlined in Section 6. The experimental and theoretical systematic uncertainties, detailed in Section 7, are taken into account for the statistical interpretation of the data. The final results are presented in Section 8. Concluding remarks are given in Section 9.

2 The ATLAS detector

The ATLAS experiment [20] at the LHC is a multi-purpose particle detector with a forward-backward symmetric cylindrical geometry¹ and a near 4π coverage in solid angle. It consists of an inner tracking detector (ID) surrounded by a thin superconducting solenoid, which provides a 2 T axial magnetic field, electromagnetic (EM) and hadron calorimeters, and a muon spectrometer (MS). The inner tracking detector covers the pseudorapidity range $|\eta| < 2.5$. It consists of silicon pixel, including the newly installed insertable B-layer [21], silicon micro-strip, and transition radiation tracking detectors. Lead/liquid-argon (LAr) sampling calorimeters provide electromagnetic energy measurements with high granularity. A hadron (steel/scintillator-tile) calorimeter covers the central pseudorapidity range ($|\eta| < 1.7$). The end-cap and forward regions are instrumented up to $|\eta| = 4.9$ with LAr calorimeters for both EM and hadronic energy measurements. The calorimeters are surrounded by the muon spectrometer and three large air-core toroidal superconducting magnets with eight coils each. The field integral of the toroid magnets

¹ ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z -axis along the beam pipe. The x -axis points from the IP to the centre of the LHC ring, and the y -axis points upwards. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the z -axis. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$. Angular distance is measured in units of $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$.

ranges between 2.0 and 6.0 T m across most of the detector. The muon spectrometer includes a system of precision tracking chambers and fast detectors for triggering. Events are selected using a first-level trigger implemented in custom electronics, which reduces the event rate to a maximum of 100 kHz using a subset of detector information. Software algorithms with access to the full detector information are then used in the high-level trigger to yield a recorded event rate of about 1 kHz [22].

3 Signal and background simulation

The production of the SM Higgs boson via gluon-gluon fusion (ggF), via vector boson fusion (VBF), associated with a vector boson (VH , where V is a W or a Z boson) and with a top quark pair ($t\bar{t}H$) is modelled with the POWHEG-BOX v2 Monte Carlo (MC) event generator [23–27]. For ggF, the PDF4LHC next-to-next-to-leading-order (NNLO) set of parton distribution functions (PDF) is used, while for all other production modes, the PDF4LHC next-to-leading-order (NLO) set is used [28]. The event generator is interfaced to EVTGEN v1.2.0 [29] for simulation of the bottom and charm hadron decays. The ggF Higgs boson production uses the POWHEG method for merging the NLO Higgs + jet cross section with the parton shower and the MiNLO method [30] to simultaneously achieve NLO accuracy for inclusive Higgs boson production. In a second step a reweighting procedure (NNLOPS), exploiting the Higgs boson rapidity distribution, is applied using the HNNLO program [31, 32] to achieve NNLO accuracy in the strong coupling constant α_s .

The matrix elements of the VBF, $q\bar{q} \rightarrow VH$ and $t\bar{t}H$ production mechanisms are calculated up to NLO in QCD. For VH production, the MiNLO method is used to merge 0- and 1-jet events [27, 33]. The $gg \rightarrow ZH$ contribution is modelled at leading order (LO) in QCD.

The production of a Higgs boson in association with a bottom quark pair ($b\bar{b}H$) is simulated at NLO with MADGRAPH5_AMC@NLO v2.3.3 [34], using the NNPDF23 PDF set [35], while the production in association with a single top quark (tH) is simulated at NLO with MADGRAPH5_AMC@NLO v2.3.3 (tHW) and with MADGRAPH5 v2.3.3 (tHq), using the CT10nlo PDF set [36].

For all production mechanisms, the PYTHIA 8 [37] generator, using the AZNLO set of tuned parameter [38], is used for the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay as well as for the parton shower modelling. All signal samples are simulated for a Higgs boson mass $m_H = 125$ GeV.

For additional cross checks, the ggF sample was also generated with MADGRAPH5_AMC@NLO. This simulation is accurate at NLO QCD accuracy for zero, one and two additional partons merged with the FxFx merging scheme [39, 40].

The Higgs boson production cross sections and decay branching ratios, as well as their uncertainties, are taken from Refs. [35, 41–48]. The ggF production is calculated with next-to-next-to-next-to-leading order (N³LO) accuracy in QCD and has NLO electroweak (EW) corrections applied [49–55]. For VBF production, full NLO QCD and EW calculations are used with approximate NNLO QCD corrections [56, 57]. The qq - and qg -initiated VH production is calculated at NNLO in QCD and NLO EW corrections are applied [58–60], while gg -initiated VH production is calculated at NLO in QCD. The $t\bar{t}H$ [61–64], $b\bar{b}H$ [65–67] and tH [68] processes are calculated to NLO accuracy in QCD. The branching ratio for the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay with $m_H = 125$ GeV is predicted to be 0.0124% [45, 69] in the SM using PROPHECY4F [70, 71], which includes the complete NLO QCD and EW corrections, and the interference effects between identical final-state fermions. Due to the latter, the expected branching ratios of the $4e$ and 4μ final states are about 10% higher than the branching ratios to $2e2\mu$ and $2\mu2e$ final states. Table 1

summarises the predicted SM production cross sections and branching ratios for the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay for $m_H = 125$ GeV.

Table 1: The predicted SM Higgs boson production cross sections (σ) for ggF, VBF and associated production with a W or Z boson or with a $t\bar{t}$ or $b\bar{b}$ pair, or with a single top quark in pp collisions for $m_H = 125$ GeV at $\sqrt{s} = 13$ TeV [28, 35, 41–43, 45–67, 69–71]. The quoted uncertainties correspond to the total theoretical systematic uncertainties calculated by adding in quadrature the uncertainties due to missing higher order corrections and PDF+ α_s . The decay branching ratio ($\mathcal{B}$) with the associated uncertainty for $H \rightarrow ZZ^*$ and $H \rightarrow ZZ^* \rightarrow 4\ell$, with $\ell = e, \mu$, is also given.

Production process	σ [pb]
ggF ($gg \rightarrow H$)	48.6 ± 2.4
VBF ($qq' \rightarrow Hqq'$)	3.78 ± 0.08
WH ($q\bar{q}' \rightarrow WH$)	1.373 ± 0.028
ZH ($q\bar{q}/gg \rightarrow ZH$)	0.88 ± 0.04
$t\bar{t}H$ ($q\bar{q}/gg \rightarrow t\bar{t}H$)	0.51 ± 0.05
$b\bar{b}H$ ($q\bar{q}/gg \rightarrow b\bar{b}H$)	0.49 ± 0.12
tH ($q\bar{q}/gg \rightarrow tH$)	0.09 ± 0.01
Decay process	$\mathcal{B} [\cdot 10^{-4}]$
$H \rightarrow ZZ^*$	262 ± 6
$H \rightarrow ZZ^* \rightarrow 4\ell$	1.240 ± 0.027

The ZZ^* continuum background from quark–antiquark annihilation is modelled using SHERPA 2.2.2 [72–74], which provides a matrix element calculation accurate to NLO in α_s for 0-, and 1-jet final states and LO accuracy for 2- and 3-jet final states. The merging with the SHERPA parton shower [75] is performed using the ME+PS@NLO prescription [76]. The NLO EW corrections are applied as a function of the invariant mass of the ZZ^* system m_{ZZ^*} [77, 78].

The gluon-induced ZZ^* production is modelled by GG2VV [79] at LO in QCD. The higher-order QCD effects for the $gg \rightarrow ZZ^*$ continuum production have been calculated for massless quark loops [80–82] in the heavy top-quark approximation [83], including the $gg \rightarrow H^* \rightarrow ZZ$ processes [84, 85]. The LO $gg \rightarrow ZZ$ simulation is scaled by the K -factor of 1.7 ± 1.0 , defined as the ratio of the higher-order and the leading-order cross section predictions.

The WZ background is modelled using POWHEG-BOX v2 interfaced to PYTHIA 8 and EVTGEN v1.2.0 for the simulation of bottom and charm hadron decays. The triboson backgrounds ZZZ , WZZ , and WWZ with four or more prompt leptons are modelled using SHERPA 2.2.2. The simulation of $t\bar{t} + Z$ events with both top quarks decaying semi-leptonically and the Z boson decaying leptonically is performed with MADGRAPH5_AMC@NLO interfaced to PYTHIA 8 and the total cross section is normalised to the prediction which includes the two dominant terms at both the LO and the NLO in a mixed perturbative expansion in the QCD and EW couplings [64]. For modelling comparisons, SHERPA 2.2.1 was used to simulate $t\bar{t} + Z$ events at LO.

The modelling of events containing Z bosons with associated jets (Z +jets) is performed using the SHERPA 2.2.1 generator. Matrix elements are calculated for up to two partons at NLO and four partons at LO using COMIX [73] and OPENLOOPS [74], and merged with the SHERPA parton shower [75] using the ME+PS@NLO prescription [76]. The NNPDF3.0 NNLO PDF set is used in conjunction with dedicated parton shower parameters tuning.

The $t\bar{t}$ background is modelled using POWHEG-BOX v2 interfaced to PYTHIA 8 for parton showering, hadronization, and the underlying event, and to EVTGEN v1.2.0 for heavy flavoured hadron decays. For this sample, the A14 parameter set [86, 87] is used. Simulated Z+jets and $t\bar{t}$ background samples are normalised to the data-driven estimate described in Section 5.

Generated events are processed through the ATLAS detector simulation [88] within the GEANT4 framework [89] and reconstructed in the same way as the data. Additional pp interactions in the same and nearby bunch crossings are included in the simulation. The pile-up events are generated using PYTHIA 8 with the A2 set of tuned parameters [90] and the MSTW2008LO PDF set [91]. The simulation samples are weighted to reproduce the observed distribution of the mean number of interactions per bunch crossing in the data.

4 Event selection

4.1 Event reconstruction

The selection and categorisation of the Higgs boson candidate events rely on the reconstruction and identification of electrons, muons and jets, closely following the analyses reported in Refs. [11, 12].

Proton-proton collision vertices are reconstructed from ID tracks with transverse momentum $p_T > 400$ MeV. The vertex with the highest $\sum p_T^2$ of reconstructed tracks is selected as the primary vertex of the hard interaction. Events are required to have at least one collision vertex with at least two associated tracks. The data are subjected to quality requirements to reject events in which detector components were not operating correctly.

Electron candidates are reconstructed from ID tracks that are matched to energy clusters in the electromagnetic calorimeter [19, 92]. A Gaussian-sum filter algorithm [93, 94] is used to compensate for radiative energy losses in the ID, while a dynamical topological cell clustering-based algorithm is used to improve the resolution of the energy clusters [19]. Electron identification is based on a likelihood discriminant combining the measured track properties, electromagnetic shower shapes and quality of the track-cluster matching. The “loose” likelihood criteria, applied in combination with track hit requirements, provide an electron reconstruction and identification efficiency of at least 90% for electrons with $p_T > 30$ GeV. Electrons are required to have $p_T > 7$ GeV and pseudorapidity $|\eta| < 2.47$, with their energy calibrated as described in Refs. [95, 96].

Muon candidate reconstruction within $|\eta| < 2.5$ is primarily performed by a global fit of fully reconstructed tracks in the ID and the MS, and a “loose” identification criterion is applied [97]. In the central detector region ($|\eta| < 0.1$), which has a reduced MS geometrical coverage, muons are also identified by matching a fully reconstructed ID track to either an MS track segment (segment-tagged muons) or a calorimetric energy deposit consistent with a minimum-ionising particle (calorimeter-tagged muons). For these two cases, the muon momentum is measured from the ID track alone. In the forward MS region ($2.5 < |\eta| < 2.7$), outside the ID coverage, MS tracks with hits in the three MS layers are accepted and combined with forward ID tracklets, if they exist (stand-alone muons). Calorimeter-tagged muons are required to have $p_T > 15$ GeV. For all other muon candidates, the minimum transverse momentum is required to be greater than 5 GeV. At most one calorimeter-tagged or stand-alone muon is allowed per event.

Muons and electrons are required to have a longitudinal impact parameter less than 0.5 mm. Additionally, muons with transverse impact parameter greater than 1 mm are rejected.

Jets are reconstructed from noise-suppressed topological clusters [98] in the calorimeter using the anti- k_t algorithm [99] with a radius parameter $R = 0.4$. The jet four-momentum is corrected for the calorimeter non-compensating response, signal losses due to noise threshold effects, energy lost in non-instrumented regions, and contributions from pile-up [100, 101]. Jets are required to have $p_T > 30$ GeV and $|\eta| < 4.5$. Jets from pile-up are suppressed using two jet-vertex-tagger multivariate discriminants, one for $|\eta| < 2.5$ [102, 103] and the other for $2.5 < |\eta| < 4.5$ [104]. Jets with $|\eta| < 2.5$ containing b -hadrons are identified using the MV2c10 b -tagging algorithm [105, 106] at an operating point with 70% b -tagging efficiency.

Ambiguities are resolved if electron, muon or jet candidates are reconstructed from the same detector information. If a reconstructed electron and muon share the same ID track, the muon is rejected if it is calorimeter-tagged; otherwise the electron is rejected. Reconstructed jets geometrically overlapping in a cone of radius $\Delta R = 0.1$ (0.2) with muons or electrons are also removed.

The missing transverse momentum, $\vec{E}_T^{\text{miss}}$, is defined as the negative vector sum of the transverse momenta of all identified and calibrated leptons and jets and remaining unclustered energy, the latter of which is estimated from low- p_T tracks associated with the primary vertex but not assigned to any lepton or jet candidate [107]. The missing transverse energy (E_T^{miss}) is defined as the magnitude of $\vec{E}_T^{\text{miss}}$.

4.2 Selection of the Higgs boson candidates

Events are triggered by a combination of unprescaled single-lepton, dilepton and trilepton triggers with different transverse momentum thresholds. Due to an increasing peak luminosity, these thresholds increase slightly during the data-taking periods [108, 109]. For single muon triggers, the p_T threshold ranges between 20 and 26 GeV, while for single electrons triggers, the p_T threshold ranges from 24 to 26 GeV. The lowest-threshold triggers are complemented by triggers with higher thresholds but looser lepton selection criteria. The global trigger efficiency for signal events passing the final selection is 98%.

In the analysis, at least two same-flavour and opposite-charge lepton pairs (hereafter referred to as lepton pairs) are required in the final state, resulting in one or more possible lepton quadruplets in each event. The three highest- p_T leptons in each quadruplet must have transverse momenta above 20 GeV, 15 GeV and 10 GeV, respectively. The lepton pair with the invariant mass m_{12} (m_{34}) closest (second closest) to the Z boson mass in each quadruplet is referred to as the leading (subleading) lepton pair. Based on the lepton flavour, each quadruplet is classified into one of the following decay channels: 4μ , $2e2\mu$, $2\mu2e$ and $4e$, with the first two leptons always representing the leading lepton pair. In each of these channels, the quadruplet with m_{12} closest to the Z boson mass has priority for selection. However all quadruplets passing the full selection are kept and if an extra lepton is found in the event a matrix-element-based pairing is used to choose the best quadruplet, as described below.

The leading lepton pair must satisfy $50 \text{ GeV} < m_{12} < 106 \text{ GeV}$. The subleading lepton pair is required to have a mass $m_{\min} < m_{34} < 115 \text{ GeV}$, where $m_{\min}$ is 12 GeV for the four-lepton invariant mass $m_{4\ell}$ below 140 GeV, rising linearly to 50 GeV at $m_{4\ell} = 190 \text{ GeV}$ and then remaining at 50 GeV for all higher $m_{4\ell}$ values. In the $4e$ and 4μ channels, the two alternative opposite-charge lepton pairings within a quadruplet must have a dilepton mass above 5 GeV to suppress the J/ψ background. The two lepton pairs within the quadruplet must have an angular separation of $\Delta R = 0.1$ (0.2) for same-flavour (different-flavour) lepton pairs. Each electron (muon) track must have a transverse impact parameter significance ($|d_0|/\sigma(d_0)$) below 5 (3) with respect to the primary vertex, to suppress the background from heavy-flavour hadrons. Reducible background from the Z +jets and $t\bar{t}$ processes is further suppressed by imposing track-based

and calorimeter-based isolation criteria on each lepton. The scalar sum of the p_T of the tracks lying within a cone of $\Delta R = 0.3$ (0.2) around the muon (electron) is required to be smaller than 15% of the lepton p_T (E_T). Similarly, the sum of the positive-energy topological clusters in a cone of $\Delta R = 0.2$ around the muon (electron) is required to be less than 30% (20%) of the lepton p_T . The calorimeter-based isolation requirement is applied after correcting for the pile-up and underlying-event contributions, as well as removing the energy deposits from the remaining three leptons. The combined signal efficiency of track-based and calorimeter-based isolation criteria is greater than 80%. Additionally, the four quadruplet leptons are required to originate from a common vertex point. A requirement corresponding to a signal efficiency of 99.5% is imposed on the χ^2 value from the fit of the four lepton tracks to their common vertex. If there is more than one decay channel per event with a quadruplet satisfying the above selection criteria, the quadruplet from the channel with highest efficiency is chosen as the Higgs boson candidate.

In case of VH or $t\bar{t}H$ production, there may be additional leptons present in the event, together with the selected quadruplet. Therefore, there is a possibility that one or more of the quadruplet leptons do not originate from a Higgs boson decay, but rather from the V boson or top quark leptonic decays. To improve the lepton pairing in such cases, a matrix-element-based pairing method is used for all events containing at least one additional lepton with $p_T > 12$ GeV and which satisfies the same identification and isolation criteria as the four quadruplet leptons [11, 12]. For all possible quadruplet combinations which pass the above selection, a matrix element (ME) for the Higgs boson decay is computed at LO using the MADGRAPH5_AMC@NLO [39] generator. The quadruplet with the largest matrix element value is selected as the Higgs boson candidate.

In order to improve the four-lepton invariant mass reconstruction, the reconstructed final-state radiation (FSR) photons in Z boson decays are accounted for using the same strategy as the previous publications [11, 12]. The Higgs boson candidates within a mass window of $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$ are selected.

The MC signal selection efficiencies in the fiducial region $|y_H| < 2.5$, where y_H is the Higgs boson rapidity, are 31%, 24%, 18% and 16%, in the 4μ , $2e2\mu$, $2\mu2e$ and $4e$ channels, respectively.

5 Background estimate

Non-resonant SM ZZ^* production via $q\bar{q}$ annihilation and gluon–gluon fusion can result in four prompt leptons in the final state and constitutes the largest background for this analysis. It is estimated using the SHERPA and GG2VV simulated samples presented in Section 3. To cross-check the theoretical modelling of this background, a ZZ^* -enriched control region is formed using the full event selection, but requiring that the four-lepton invariant mass not lie within the region $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$. Similarly, contribution from $t\bar{t} + Z$ and VVV are estimated and cross-checked using the MADGRAPH5_AMC@NLO and SHERPA simulated samples presented in Section 3.

Other processes that contribute to the background, such as $Z + \text{jets}$, $t\bar{t}$, and WZ , contain at least one jet, photon or lepton from a hadron decay that is misidentified as a prompt lepton. These backgrounds are significantly smaller than the non-resonant ZZ^* irreducible background and are estimated using data where possible, following slightly different approaches for the $\ell\ell + \mu\mu$ and $\ell\ell + ee$ final states [11, 12].

In the $\ell\ell + \mu\mu$ final states, the normalisations for the $Z + \text{jets}$ and $t\bar{t}$ backgrounds are determined performing fits to the invariant mass of the leading lepton pair in dedicated independent control regions. The control regions are formed by relaxing the χ^2 requirement on the four-lepton vertex fit, and by inverting or relaxing isolation and/or impact-parameter requirements on the subleading muon pair. Additional control regions

($e\mu\mu\mu$ and $\ell\ell + \mu^\pm\mu^\pm$) are used to improve the background estimate by reducing the statistical error on the fitted normalisations. Transfer factors to extrapolate from the control regions to the signal region are obtained separately for $t\bar{t}$ and $Z + \text{jets}$ using simulation. The shapes of the $Z + \text{jets}$ and $t\bar{t}$ backgrounds for the differential observables and categories are taken from simulation and normalised using the inclusive data-driven estimate.

The $\ell\ell + ee$ control-region selection requires the electrons in the subleading lepton pair to have the same charge, and relaxes the identification and isolation requirements on the electron candidate with the lowest transverse energy. This electron candidate, denoted as X , can be a light-flavour jet, an electron from photon conversion or an electron from heavy-flavour hadron decay. The heavy-flavour background is completely determined from simulation, whereas the light-flavour and photon conversion background is obtained with the sPlot [110] method. This is based on a fit to the number of hits in the innermost ID layer in the data control region. Transfer factors to extrapolate from the $\ell\ell + ee$ control-region to the signal region for the light-flavour jets and converted photons, obtained from simulated samples, are corrected using a $Z + X$ control region and then used to extrapolate the extracted yields to the signal region. Both the extraction of the yield in the control region and the extrapolation are performed in bins of the transverse momentum of the electron candidate and the jet multiplicity. In order to extract the shape of the backgrounds from light-flavour jets and photon conversions in each observable, a similar method is used, except that the extraction and extrapolation is performed as a function of the transverse momentum of the electron candidate in each variable of interest.

6 Analysis strategy

6.1 Inclusive and differential cross sections

Fiducial production cross sections are measured both inclusively and separately for each of the final states of the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay ($4\mu, 2e2\mu, 2\mu2e, 4e$). Differential fiducial cross sections are presented for two observables that describe Higgs boson production in pp collisions. They are inclusive in the different final states and Higgs boson production processes. The Higgs boson transverse momentum $p_{T,4\ell}$ can be used to test perturbative QCD calculations and it is also sensitive to the Lagrangian structure of the Higgs boson interactions [111]. The number of jets, N_{jets} , produced in association with the Higgs boson provides sensitivity to the theoretical modelling of gluon emission and the fractions of the different production modes.

6.1.1 Fiducial phase space

The fiducial cross sections are defined at particle level using the selection requirements outlined in Table 2, which are chosen to closely match those in the detector-level analysis in order to minimise model-dependent acceptance extrapolations.

The fiducial selection is applied to final-state² electrons and muons that do not originate from hadrons or τ decays, after “dressing” them, i.e. the four-momenta of photons within a cone of size $\Delta R = 0.1$ are added to the lepton four-momentum. The photons are required to not originate from hadron decays. Particle-level

² Final-state particles are defined as particles with a lifetime $c\tau > 10$ mm. For electrons and muons, this corresponds to leptons after final state radiation.

Table 2: List of event selection requirements which define the fiducial phase space for the cross section measurement. SFOS lepton pairs are same-flavour opposite-sign lepton pairs.

Leptons and jets	
Leptons:	$p_T > 5 \text{ GeV}, \eta < 2.7$
Jets:	$p_T > 30 \text{ GeV}, y < 4.4$
remove jets with:	$\Delta R(\text{jet}, \ell) < 0.1$
Lepton selection and pairing	
Lepton kinematics:	$p_T > 20, 15, 10 \text{ GeV}$
Leading pair (m_{12}):	SFOS lepton pair with smallest $ m_Z - m_{\ell\ell} $
Subleading pair (m_{34}):	remaining SFOS lepton pair with smallest $ m_Z - m_{\ell\ell} $
Event selection (at most one quadruplet per event)	
Mass requirements:	$50 \text{ GeV} < m_{12} < 106 \text{ GeV}$ and $12 \text{ GeV} < m_{34} < 115 \text{ GeV}$
Lepton separation:	$\Delta R(\ell_i, \ell_j) > 0.1$
J/ψ veto:	$m(\ell_i, \ell_j) > 5 \text{ GeV}$ for all SFOS lepton pairs
Mass window:	$115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$
If extra leptons with $p_T > 12 \text{ GeV}$:	Quadruplet with the largest ME

jets are reconstructed from final-state particles using the anti- k_t algorithm with radius parameter $R = 0.4$. Electrons, muons, neutrinos (if they are not from hadron decays) and photons used to dress leptons, are excluded from the jet clustering. Jets are removed if they are within a cone of size $\Delta R = 0.1$ around a selected lepton.

Using the same procedure as for reconstructed events, the selected dressed leptons are used to form quadruplets. In case of VH or $t\bar{t}H$ production, additional leptons not originating from a Higgs boson decay can induce a “lepton mispairing” when assigning them to the leading and subleading Z bosons. To improve the lepton pairing efficiency, the matrix-element-based pairing method as described in Section 4.2 is employed.

The variables used in the differential cross section measurement are calculated using the dressed leptons of the quadruplets.

The acceptance of the fiducial selection, defined as the ratio of the number of events passing the particle-level selection and the events generated in a given bin or channel, (with respect to the full phase space of $H \rightarrow ZZ^* \rightarrow 2\ell 2\ell'$, where $\ell, \ell' = e$ or μ) is 49% for a SM Higgs boson with $m_H = 125 \text{ GeV}$. The ratio C of the number of events passing the selection after detector simulation and event reconstruction to those passing the particle-level selection is 42%. Due to resolution effects, about 1.6% of the events which pass the detector-level selection fail the particle-level selection. The acceptance and correction factors depend on the event topology, and they vary from 40% to 60% for the highest $p_{T,4\ell}$ and N_{jets} bins due to the higher boost of the four lepton system.

6.1.2 Signal extraction and correction factors

To extract the number of signal events in each bin of a differential distribution (or for each decay channel for the inclusive fiducial cross section), invariant mass templates for the Higgs boson signal and the background processes are fit to the $m_{4\ell}$ distribution in data. The overall normalisation of the backgrounds

can vary within the systematic uncertainties. For the differential distributions, no split into decay channels is performed, and the SM $ZZ^* \rightarrow 4\ell$ decay fractions are assumed.

The fiducial cross section $\sigma_{i,\text{fid}}$ for a given final state or bin of the differential distribution, i , is defined as:

$$\sigma_{i,\text{fid}} = \sigma_i \times A_i \times \mathcal{B} = \frac{N_{i,\text{fit}}}{\mathcal{L} \times C_i}, \quad C_i = \frac{N_{i,\text{reco}}}{N_{i,\text{part}}}, \quad (1)$$

where A_i is the acceptance in the fiducial phase space, σ_i is the total cross section in bin i and $\mathcal{B}$ is the branching ratio. The term $N_{i,\text{fit}}$ is the number of extracted signal events in data, $\mathcal{L}$ is the integrated luminosity and C_i is the bin-by-bin correction factor for detector efficiency and resolution. The term $N_{i,\text{reco}}$ is the number of reconstructed signal events and $N_{i,\text{part}}$ is the number of events at the particle level in the fiducial phase-space. The inclusive fiducial and differential cross sections are measured using a binned profile-likelihood-ratio fit as described in Section 8.1. The correction factor is calculated from simulated Higgs boson samples, assuming SM production mode fractions and $ZZ^* \rightarrow 4\ell$ decay fractions as discussed in Section 3. The bin-by-bin migration is less than 25% in all the bins of the Higgs boson transverse momentum distribution, and less than 30% for the bins of the N_{jets} distribution. The experimental systematic uncertainties of the correction factors and signal shapes are described in Section 7. The correction factors for the different Higgs boson production modes agree within 15%, except for the $t\bar{t}H$ mode, which differs by up to 40%. This is due to $t\bar{t}H$ events having more hadronic jets and no isolation requirements being applied to the leptons at the particle level.

6.2 Production mode cross sections

The Higgs boson couplings to heavy SM vector bosons (W and Z), top quarks and gluons are studied by measuring the cross sections for different production processes. To achieve this, the reconstructed Higgs boson candidate events are classified into eleven categories, each sensitive to a different Higgs boson production mechanism or kinematic region. For simplicity, these regions are called “production bins”. The event yields in each category serves as observable. To improve the sensitivity, boosted decision trees (BDTs) [112] based discriminating observables are introduced in a subset of the categories.

6.2.1 Simplified template cross sections

Theoretical uncertainties have a reduced impact on $\sigma \cdot \mathcal{B}(H \rightarrow ZZ^*)$ results and enter primarily for the interpretation of the results in terms of Higgs boson couplings. The definitions of the production bins shown in the left hand-side panel of Figure 1 (shaded area) are based on particle-level events produced by dedicated event generators closely following the framework of *Simplified Template Cross Sections* [44]. The bins are chosen in such a way that the measurement precision is maximised and at the same time possible Beyond Standard Model (BSM) contributions can be probed. All production bins are defined for Higgs bosons with rapidity $|y_H| < 2.5$ and no requirements are placed on the particle-level leptons. Two sets of production bins are considered, with different granularity, as a different trade-off between statistical and theoretical uncertainties.

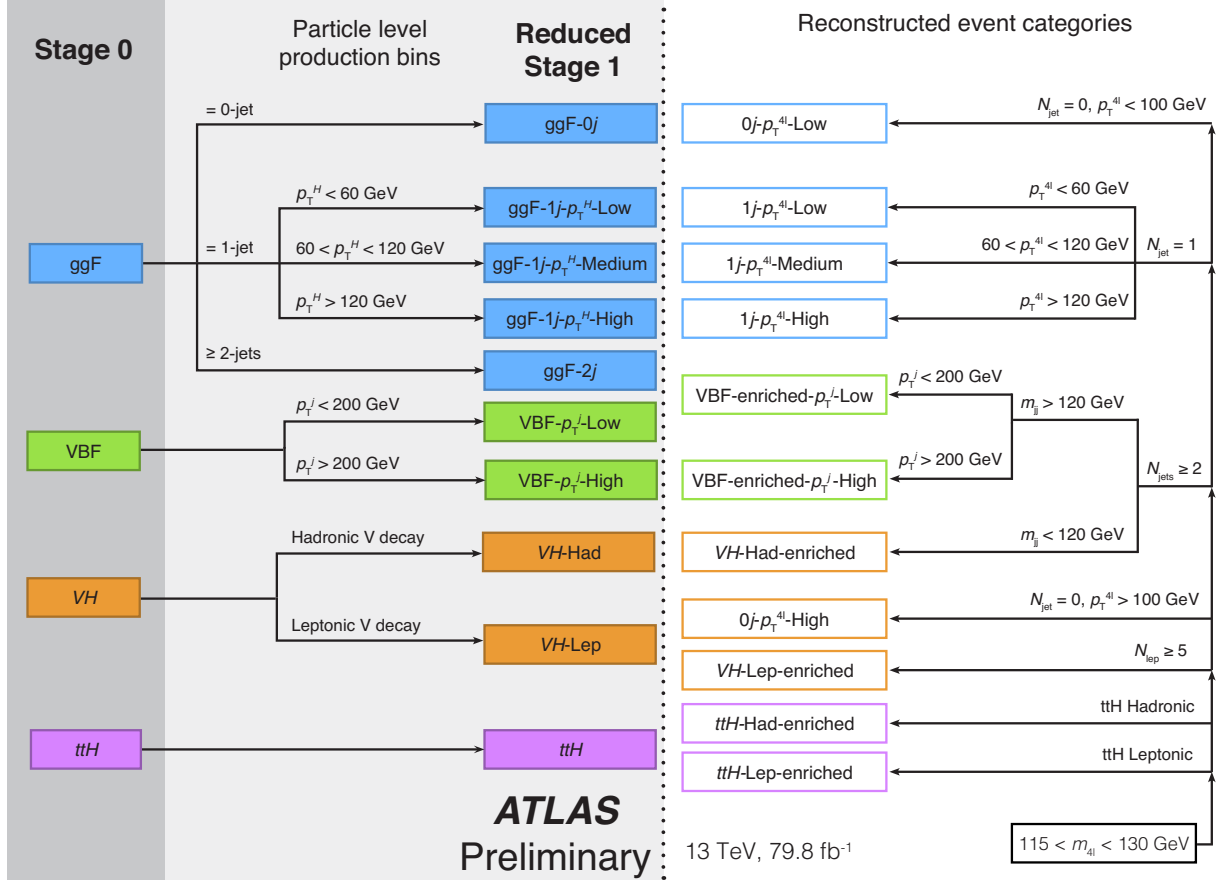


Figure 1: The phase-space regions (production bins) for the measurement of the Higgs boson production cross sections which are defined at the particle level for Stage 0 and 1, and the corresponding reconstructed event categories. The description of production bins is given in Section 6.2.1, while the reconstructed event categories are described in Section 6.2.2. The bbH (tH) contribution is included in the ggF (ttH) production bins.

For the first set (Stage 0) [44], production bins are simply defined according to the Higgs boson production mode: gluon–gluon fusion, vector boson fusion and associated production with top quark pairs or vector bosons. The bbH Higgs boson production bin is not included because there is insufficient sensitivity to measure this process with the current integrated luminosity. The bbH and ggF production modes have similar acceptance, their contributions are therefore considered together in the analysis, and their relative ratio is fixed to the SM prediction. The sum of their contributions is referred to in the following as gluon–gluon fusion. Similarly, single top production (tH) is considered together with ttH , their relative ratio is fixed to the SM prediction. Differently from Stage 0 described in Ref. [44], the VH events with a hadronically decaying vector boson V are not included in the VBF bin.

For the second set (reduced Stage 1), a more exclusive group of production bins is defined. This set is obtained by the merging of those production bins of the original Stage-1 set from Ref. [44], which cannot be measured separately in the $H \rightarrow ZZ^* \rightarrow 4\ell$ channel with the current data sample. These bins are predominately defined using the Higgs boson transverse momentum (p_T^H) and particle-level jets, which are built from all stable particles (all particles with $c\tau > 10$ mm) including neutrinos, photons and

leptons from hadron decays or produced in the shower. All Higgs boson decay products, as well as the leptons and neutrinos from the decays of the signal V bosons are removed, while the decay products from hadronically decaying signal V bosons are included in the inputs to the particle-level jet building. The anti- k_t jet reconstruction algorithm, implemented in the FastJet package [113], with a radius parameter $R = 0.4$ is used and jets are required to have $p_T > 30$ GeV. The gluon–gluon fusion process is split into events with zero, one or at least two particle-level jets. The 1-jet bin is further split into three bins with the Higgs boson transverse momentum below 60 GeV, between 60 GeV and 120 GeV, and above 120 GeV. The reduced Stage-1 gluon–gluon fusion bins are correspondingly denoted by ggF-0j, ggF-1j- p_T^H -Low, ggF-1j- p_T^H -Med, ggF-1j- p_T^H -High and ggF-2j. The VBF production bin is split into two bins with the transverse momentum of the leading jet, p_T^{j1} , below and above 200 GeV (VBF- p_T^j -Low and VBF- p_T^j -High, respectively). The former bin is expected to be dominated by SM events, while the latter is sensitive to potential BSM contributions. For VH production, separate bins with hadronically (VH -Had) and leptonically (VH -Lep) decaying vector bosons are considered. The leptonic V boson decays include the decays into τ leptons and into neutrino pairs. Differently from Stage 1 described in Ref. [44] the VH -Had is not included in the VBF. The $t\bar{t}H$ production bin remains the same as for Stage 0.

The right hand-side panel of Figure 1 summarises the corresponding categories of reconstructed events in which the cross section measurements are performed and which are described in more detail in Section 6.2.2. There is a dedicated reconstructed event category for each production bin except for ggF-2j. This process contributes strongly to all reconstructed event categories containing events with at least two jets, and can therefore be measured in these categories, with the highest sensitivity expected in the VBF-enriched- p_T^j -Low category.

6.2.2 Categorisation of reconstructed Higgs boson event candidates

The classification of events is performed in the following order. First, events are classified as enriched in the $t\bar{t}H$ process ($t\bar{t}H$ -enriched) by requiring at least one b -tagged jet in the event (with 70% b -tagging efficiency). The $t\bar{t}H$ -enriched category is split according to the decay mode of the two W bosons from the top quark decays. For the fully hadronic decay ($t\bar{t}H$ -Had-enriched), there must be at least four additional jets, while for semi- and di-leptonic decays ($t\bar{t}H$ -Lep-enriched), at least one additional lepton with $p_T > 12$ GeV³ together with at least two jets is required. Events with additional leptons but not satisfying the above jet requirements define the next category enriched in VH production with leptonic vector boson decays (VH -Lep-enriched).

The remaining events are classified according to their jet multiplicity into events with no jets, exactly one jet or at least two jets. Among events with at least two jets there are significant contributions from the VBF and VH production modes in addition to ggF. These events are divided into two categories according to the invariant mass m_{jj} of the two leading jets. The requirement of $m_{jj} \leq 120$ GeV enhances the VH production mode with hadronically decaying vector bosons (VH -Had-enriched). For $m_{jj} > 120$ GeV, the VBF Higgs boson signal is dominant, and these events are further classified according to the transverse momentum of the leading jet into events with p_T^{j1} below (VBF-enriched- p_T^j -Low) and above 200 GeV (VBF-enriched- p_T^j -High). Events with zero or one jet in the final state are expected to be dominated by the ggF process. Following the particle-level definition of production bins from Section 6.2.1, the 1-jet category is further split into three categories with the four-lepton transverse momentum $p_T^{4\ell}$ smaller than

³ The additional lepton is required to satisfy the same identification, isolation, impact parameter and angular separation requirements as the leptons in the quadruplet.

60 GeV ($1j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$), between 60 and 120 GeV ($1j\text{-}p_{\text{T}}^{4\ell}\text{-Med}$), and larger than 120 GeV ($1j\text{-}p_{\text{T}}^{4\ell}\text{-High}$). The largest number of ggF events and the highest ggF purity are expected in the zero-jet category ($0j$). The zero-jet category is split in two categories with $p_{\text{T}}^{4\ell}$ smaller ($0j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$) and larger than 100 GeV ($0j\text{-}p_{\text{T}}^{4\ell}\text{-High}$). These additional categories improve the discrimination between VH ($V \rightarrow \ell\nu/\nu\nu$) and ggF.

The expected number of signal events is shown in Table 3 for each Stage-0 production bin and separately for each reconstructed event category. The ggF and bbH contributions are shown separately in order to compare their relative contributions, but both belong in the same (ggF) production bin. The highest bbH event yield is expected in the $0j$ category since the jets tend to be more forward than in the $t\bar{t}H$ process, thus escaping the acceptance of the $t\bar{t}H$ selection criteria. The sources of uncertainty on these expectations are detailed in Section 7. The signal composition in terms of the reduced Stage-1 production bins is shown in Figure 2. The separation of the contributions from different production bins, such as the sizeable contribution of the ggF- $2j$ component in reconstructed categories with two or more jets, is further improved by means of boosted decision tree observables, as described in the following.

Table 3: The expected number of SM Higgs boson events with a mass $m_H = 125$ GeV in the mass range $115 < m_{4\ell} < 130$ GeV for an integrated luminosity of 79.8 fb^{-1} and $\sqrt{s} = 13$ TeV in each reconstructed event category, shown separately for each Stage-0 production bin. The ggF and bbH yield are shown separately but both contribute to the same (ggF) production bin. ZH and WH ($t\bar{t}H$ and tH) are reported separately but they are merged together for the final result. Statistical and systematic uncertainties, including the theory ones, are added in quadrature. The symbol - represents the cases in which the yield is < 0.001 . Compared to Ref. [114], updated jet uncertainties with smaller impact are used.

Reconstructed event category	SM Higgs boson production mode						
	ggF	VBF	WH	ZH	$t\bar{t}H$	bbH	tH
$0j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$	54 ± 5	0.64 ± 0.12	0.213 ± 0.032	0.199 ± 0.030	-	0.56 ± 0.28	-
$1j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$	16.1 ± 2.2	1.05 ± 0.06	0.291 ± 0.035	0.173 ± 0.021	0.0017 ± 0.0010	0.23 ± 0.12	0.00140 ± 0.00030
$1j\text{-}p_{\text{T}}^{4\ell}\text{-Med}$	9.6 ± 1.5	1.38 ± 0.15	0.292 ± 0.033	0.194 ± 0.022	0.0018 ± 0.0011	0.049 ± 0.025	0.0021 ± 0.0004
$1j\text{-}p_{\text{T}}^{4\ell}\text{-High}$	2.4 ± 0.5	0.60 ± 0.07	0.115 ± 0.014	0.106 ± 0.013	0.0018 ± 0.0006	0.009 ± 0.004	0.0017 ± 0.0004
VBF-enriched- $p_{\text{T}}^{4\ell}\text{-Low}$	7.8 ± 1.6	4.1 ± 0.4	0.35 ± 0.05	0.29 ± 0.04	0.124 ± 0.013	0.10 ± 0.05	0.055 ± 0.007
VBF-enriched- $p_{\text{T}}^{4\ell}\text{-High}$	5.5 ± 1.1	0.43 ± 0.04	0.68 ± 0.07	0.52 ± 0.05	0.051 ± 0.008	0.053 ± 0.027	0.0169 ± 0.0022
$VH\text{-Had-enriched}$	0.70 ± 0.20	0.38 ± 0.04	0.062 ± 0.010	0.050 ± 0.008	0.038 ± 0.005	0.0014 ± 0.0007	0.0119 ± 0.0013
$VH\text{-Lep-enriched}$	0.030 ± 0.004	0.0084 ± 0.0004	0.44 ± 0.04	0.116 ± 0.011	0.083 ± 0.011	0.0028 ± 0.0014	0.0172 ± 0.0018
$0j\text{-}p_{\text{T}}^{4\ell}\text{-High}$	0.059 ± 0.022	0.0096 ± 0.0017	0.030 ± 0.004	0.085 ± 0.010	-	-	-
$t\bar{t}H\text{-Had-enriched}$	0.09 ± 0.09	0.020 ± 0.004	0.0130 ± 0.0027	0.028 ± 0.006	0.38 ± 0.04	0.012 ± 0.006	0.054 ± 0.006
$t\bar{t}H\text{-Lep-enriched}$	-	-	0.0026 ± 0.0006	0.0018 ± 0.0004	0.212 ± 0.025	-	0.0204 ± 0.0022
Total	97 ± 8	8.6 ± 0.4	2.49 ± 0.25	1.76 ± 0.17	0.90 ± 0.09	1.0 ± 0.5	0.181 ± 0.020

6.2.3 Discriminating observables

In order to further increase the sensitivity of the cross section measurements in the production bins (Section 6.2.1), BDT discriminants are introduced in reconstructed event categories in the fit procedure. The BDTs are trained on simulated SM Higgs signal and non-Higgs background or the other production processes, based on several discriminating observables as summarised in Table 4.

A BDT discriminant in the $0j\text{-}p_{\text{T}}^{4\ell}\text{-Low}$ category is built to separate the Higgs boson signal from the non-resonant ZZ^* background, relying on the four-lepton transverse momentum and rapidity, as well as on the kinematic discriminant D_{ZZ^*} [115], defined as the difference between the logarithms of the signal and background matrix elements squared. In the two 1-jet categories with $p_{\text{T}}^{4\ell}$ below 120 GeV, a BDT

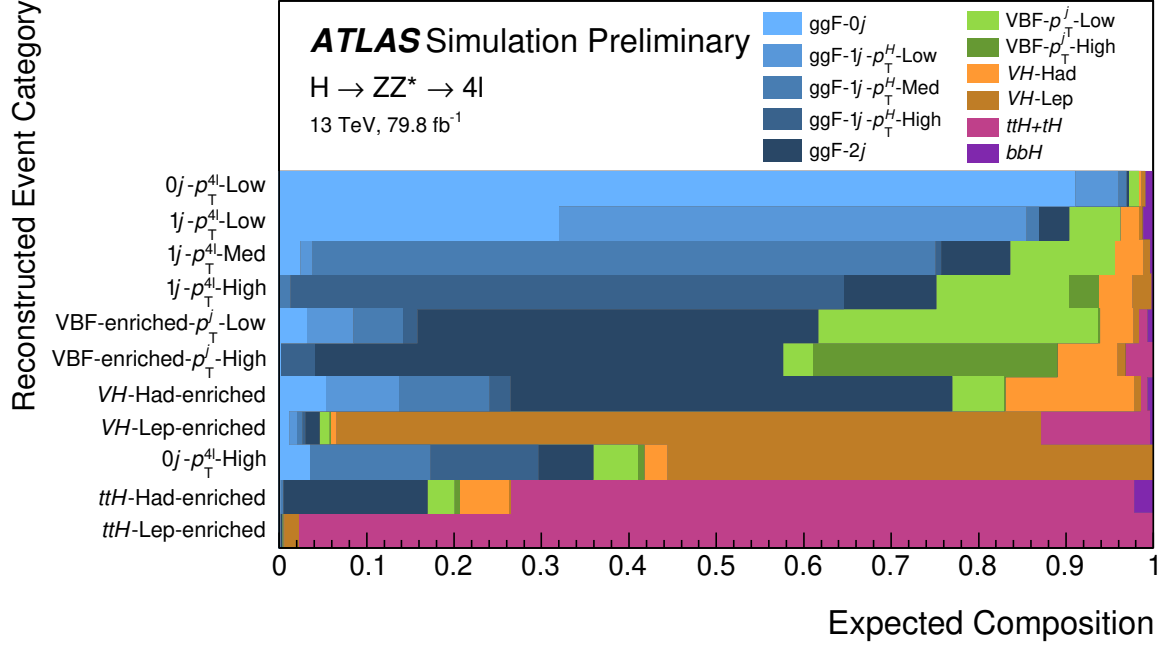


Figure 2: Standard Model signal composition in terms of the reduced Stage-1 production bins in each reconstructed event category. The ggF and bbH contributions are shown separately but both contribute to the same (ggF) production bin.

Table 4: The BDT discriminants and their corresponding input variables used for the measurement of cross sections per production bin. The jets are denoted by “ j ”. See the text for variable definitions.

Reconstructed event category	BDT discriminant	Input variables
$0j-p_T^{4\ell}$ -Low	BDT_{ggF}	$p_T^{4\ell}, \eta_{4\ell}, D_{ZZ^*}$
$1j-p_T^{4\ell}$ -Low	$\text{BDT}_{\text{VBF}}^{1j-p_T^{4\ell}\text{-Low}}$	$p_T^j, \eta_j, \Delta R(j, 4\ell)$
$1j-p_T^{4\ell}$ -Med	$\text{BDT}_{\text{VBF}}^{1j-p_T^{4\ell}\text{-Med}}$	$p_T^j, \eta_j, \Delta R(j, 4\ell)$
VBF-enriched- p_T^j -Low	BDT_{VBF}	$m_{jj}, \Delta\eta_{jj}, p_T^{j1}, p_T^{j2}, \eta_{4\ell}^*, \Delta R_{jZ}^{\min}, p_T^{4\ell jj}$
VH-Had-enriched	$\text{BDT}_{\text{VH-Had}}$	$m_{jj}, \Delta\eta_{jj}, p_T^{j1}, p_T^{j2}, \eta_{4\ell}^*, \Delta R_{jZ}^{\min}, \eta_{j1}$
ttH -Had-enriched	$\text{BDT}_{ttH\text{-Had}}$	$m_{jj}, \Delta\eta_{jj}, \Delta R_{jZ}^{\min}, \Delta R(j, 4\ell), \eta_{4\ell}^*,$ $E_T^{\text{miss}}, p_T^{jj}, N_{\text{jets}}, N_{b\text{-jets}}, H_T, \mathcal{M}_{\text{sig}}$

discriminant combining information about the jet transverse momentum (p_T^j), rapidity (η_j) and angular separation between the jet and the four-lepton system ($\Delta R(j, 4\ell)$) is introduced to distinguish between ggF and VBF Higgs boson production. In the VBF-enriched- p_T^j -Low (VH -Had-enriched) category, the separation of the VBF (VH) from ggF (ggF and VBF) production mechanism is achieved by means of the following input variables: m_{jj} , pseudorapidity separation ($\Delta\eta_{jj}$) and transverse momenta of the two leading jets (p_T^{j1} and p_T^{j2}), the difference between the pseudorapidity of the four-lepton system and the average pseudorapidity of the two leading jets ($\eta_{4\ell}^*$) [116], as well as the minimum angular separation between the leading lepton pair and the two leading jets ($\Delta R_{jZ}^{\min}$). In addition, the pseudorapidity of the leading jet (η_{j1}) is used as an input in the VH -Had-enriched category, while the transverse momentum of the Higgs–dijet system, $p_T^{4\ell jj}$, is employed for the VBF-enriched category. For $p_T^{4\ell jj} < 50$ GeV it is replaced by $p_T^{4\ell jj} = 50$ GeV in order to reduce the QCD scale uncertainty. In the $t\bar{t}H$ -Had-enriched category, a BDT discriminant is introduced to separate $t\bar{t}H$ from ggF Higgs boson production and the $t\bar{t} + Z$ processes. This BDT uses the m_{jj} , $\Delta\eta_{jj}$, $\Delta R_{jZ}^{\min}$, $\Delta R(j, 4\ell)$, $\eta_{4\ell}^*$ variables. Additionally, the missing transverse momentum (E_T^{miss}), the transverse momentum p_T^{jj} of the two leading jets, the number of jets (N_{jets}), the number of b -tagged jets ($N_{b\text{-jets}}$), the scalar sum of the transverse momenta of all selected jets (H_T) and the matrix element evaluated for the Higgs boson signal process ($\mathcal{M}_{\text{sig}}$) [115] are also included in the $\text{BDT}_{t\bar{t}H\text{-Had}}$ training.

7 Systematic uncertainties

The systematic uncertainties are grouped into experimental and theoretical uncertainties. The first category includes uncertainties on lepton and jet reconstruction, identification, isolation and trigger efficiencies, energy resolution and scale, and uncertainty on the total integrated luminosity. Uncertainties from the procedure used to derive the data-driven background estimates are also included in this category. The second category includes uncertainties in theoretical modelling of the signal and background processes.

The uncertainties can affect the signal acceptance, selection efficiency, physics object reconstruction and identification efficiency, correction factors and discriminant distributions as well as the background estimates. The dominant sources of uncertainty and their effect are described in the following subsections. The impact of these uncertainties on the measurements is summarised in Table 5.

7.1 Experimental uncertainties

The uncertainty in the integrated luminosity is 2.0%, which affects the signal and simulated background estimates. It is derived, following a methodology similar to the one described in Ref. [117], from a preliminary calibration of the luminosity scale using x – y beam-separation scans.

The uncertainty in the predicted yields due to pile-up modelling ranges between 2% and 5% and is derived by varying the average number of pile-up events in the simulation to cover the uncertainty in the ratio of the predicted to measured inelastic cross sections [118].

The electron (muon) reconstruction, isolation and identification efficiencies, and the energy (momentum) scale and resolution are derived from data using large samples of $J/\psi \rightarrow \ell\ell$ and $Z \rightarrow \ell\ell$ decays [95, 97, 119]. Typical uncertainties in the predicted yield due to the identification and reconstruction efficiency uncertainties are in the range 1.0–2.0% for muons and 1.0–4.0% for electrons. The uncertainty in the

Table 5: The impact of the dominant systematic uncertainties (in percent) on the measured inclusive fiducial, and the Stage-0 production mode cross sections. Similar sources of systematic uncertainties are grouped together - Luminosity (Lum.), electron/muon reconstruction and identification efficiencies and pile-up (e , μ , tagging), jet energy scale/resolution and b -tagging efficiencies (Jets, flavour tagging), uncertainties on reducible background (Reducible backgr.), theoretical uncertainties on ZZ^* background (ZZ^* backgr.), and theoretical uncertainties on the signal due to parton density function (PDF), QCD scale and showering algorithm (Parton shower). The uncertainties on correction factor due to the relative contribution of each Higgs boson production mechanism (Signal composition) are only applicable for fiducial cross section measurements.

Measurement	Experimental uncertainties [%]				Theory uncertainties [%]				
	Lum.	e , μ , pile-up	Jets, flavour tagging	Reducible backgr.	ZZ^* backgr.	PDF	QCD scale	Signal Parton Shower	Composition
Fiducial cross section									
	2.8	4.3	< 0.1	0.3	1.6	0.6	0.5	0.4	0.1
Per decay channel fiducial cross sections									
4μ	2.8	3.9	< 0.1	0.3	1.6	0.6	0.4	0.6	0.2
$4e$	2.8	9.0	< 0.1	1.0	1.6	0.6	0.8	0.5	0.1
$2\mu 2e$	2.7	8.6	< 0.1	0.9	1.5	0.6	0.7	0.5	0.1
$2e 2\mu$	2.8	3.6	< 0.1	0.4	1.8	0.6	0.7	0.5	0.2
Stage-0 production bin cross sections									
ggF	2.9	3.9	1.3	0.7	2.3	0.4	2.1	0.7	-
VBF	1.7	1.5	10.5	0.5	2.3	2.3	9.5	5.1	-
VH	2.0	1.7	7.8	1.8	5.6	2.1	14.9	3.1	-
$t\bar{t}H$	2.5	1.9	3.9	1.5	1.9	0.3	8.8	9.6	-

expected yields coming from the muon and electron isolation efficiency uncertainties are also taken into account, with the typical size being 2%. The uncertainties in the electron and muon energy and momentum scale and resolution are small and have a negligible impact on the measurements presented in Section 8.

The uncertainties in the jet energy scale and resolution are in the range of 3–7% and 2–4%, respectively [120, 121]. Compared to Ref. [114], updated jet uncertainties with smaller impact are used. The impact of these uncertainties are more relevant for the VH , VBF and $t\bar{t}H$ production modes and N_{jets} differential cross section measurements (5–20%) and for all the reduced Stage-1 cross section measurements, including the ggF process split into the different N_{jets} exclusive production bins (5–20%). They are negligible for the inclusive, fiducial and $p_{T,4\ell}$ differential cross section measurements.

The impact from the uncertainty in the efficiency of the b -tagging algorithm, which is derived from $t\bar{t}$ events, is at the level of a few percent over most of the jet p_T range [106]. This uncertainty is only relevant in the $t\bar{t}H$ -enriched category, with its expected impact being approximately 5% in the $t\bar{t}H$ cross section measurement. The uncertainties associated with the E_T^{miss} reconstruction have a negligible impact in the $t\bar{t}H$ cross section measurement.

For the data-driven measurement of the reducible background, three sources of uncertainty are considered. As the yields are estimated using a statistical fit to a control data sample, the inclusive background estimate has a relatively small (4%) statistical uncertainty. The lepton identification, isolation and impact parameter significance efficiency is compared between data and simulated events in a separate region enriched with on-shell Z boson decays accompanied by an electron or a muon. The difference in efficiency is assigned as uncertainty on the extrapolation of the yield estimate from the control region to the signal region. The combination of these two uncertainties (6%) of the inclusive reducible background estimate is considered as correlated across the reconstructed event categories. There is an additional uncorrelated uncertainty

(8–70%) in the fraction of the reducible background in each experimental category due to the statistical precision of the simulated samples.

7.2 Theoretical uncertainties

The theoretical modelling of the signal and background processes is affected by uncertainties due to missing higher order corrections, modelling of parton showers and underlying event, and PDF+ α_s uncertainties.

The impact of the theory systematic uncertainties on the signal depends on the kind of measurement that is performed. For signal strength measurements, defined as the measured cross section divided by the SM prediction, each source of theory uncertainty affects both the fiducial acceptance and the predicted SM cross section. For the cross section measurements, only effects on the acceptance need to be considered.

One of the dominant sources of theoretical uncertainty is the prediction of the ggF process in the different N_{jets} categories. The ggF process has a large contribution in categories with at least 2 jets. To estimate the variations due to the choice of QCD scale and migration effects on the N_{jets} ggF cross sections, the approach described in Ref. [44] is used, which exploits the latest predictions for the inclusive jet cross sections. In particular, the uncertainty from the choice of the factorisation and renormalisation scales, the choice of resummation scales, and the migrations between the 0-jet and 1-jet phase-space bins or between the 1-jet and ≥ 2 -jet bins are considered. The impact of QCD scale variations on the Higgs boson p_T distribution is taken into account as an additional uncertainty. The uncertainty in the Higgs boson p_T at higher order originating from the assumption of infinite top quark mass in the heavy-quark loop is also taken into account by comparing the p_T distribution predictions to finite-mass calculations. An additional uncertainty in the acceptance of the ggF process in VBF topologies due to missing higher orders in QCD in the calculation is estimated by variations of the renormalisation and factorisation scales using fixed-order calculations with MCFM [122]. For the other production modes, the QCD scales uncertainties are obtained by varying the renormalisation and factorisation scale by factors of two. The configuration with the largest impact, the relative difference between the varied and the nominal configuration, is chosen to define the uncertainty in each experimental category. QCD scale uncertainties are treated as uncorrelated among the different production modes.

The uncertainties in the acceptances due to the modelling of parton showers and multiple-parton interactions are estimated with AZNLO tune eigenvector variations and by comparing the acceptance using the parton showering algorithm from PYTHIA 8 and from HERWIG7 [123] for the ggF, VBF and VH processes, while HERWIG++ [124] is compared with PYTHIA 8 for the $t\bar{t}H$ process. The uncertainty due to each AZNLO tune variation is taken as correlated among the different production modes while the difference between the parton showering algorithms is treated as an uncorrelated uncertainty. A 100% uncertainty is assigned to heavy flavour quark production modelling for the ggF contribution entering in the $t\bar{t}H$. This has a negligible impact on the final results.

The impact of the PDF uncertainties is estimated with the eigenvector variations of the PDF4LHC_NLO_30 Hessian PDF set following the PDF4LHC recommendations [28]. The modification of the predictions originating from each eigenvector variation is added as a separate source of uncertainty in the model. The same procedure is applied for the ggF, VBF, VH and $t\bar{t}H$ processes, enabling correlations to be taken into account in the fit model.

The impacts of the sources of theoretical uncertainty, as described above, on the shape of BDT discriminants are also considered. In addition, for VBF production, the changes in the $\Delta\eta_{jj}$ distribution as predicted

at NNLO compared to NLO in QCD [125] are considered and shown to have a negligible impact on the BDT distributions. For ggF production, a further cross-check is performed by comparing the BDT shapes in the corresponding categories as predicted by POWHEG NNLOPS and MADGRAPH5_AMC@NLO with the FxFx merging scheme. The $\text{BDT}_{\text{VBF}}^{1j-p_T^{4\ell}\text{-Low}}$, $\text{BDT}_{\text{VBF}}^{1j-p_T^{4\ell}\text{-Med}}$, BDT_{VBF} and $\text{BDT}_{VH\text{-Had}}$ shapes from the two generators agree within the statistical uncertainties and, therefore, no additional shape uncertainty is included. For $\text{BDT}_{t\bar{t}H\text{-Had}}$, a 25% shape uncertainty is applied to the highest BDT bin. Similarly, a 5% shape uncertainty is applied to $\text{BDT}_{t\bar{t}H\text{-Had}}$ to account for the ggF modelling difference as predicted by POWHEG NNLOPS and POWHEG using the HJJ-MiNLO method.

For signal strength and cross sections extrapolated to the full phase space, an additional uncertainty (2.2%) related to the $H \rightarrow ZZ^*$ branching ratio [70, 71] are included in the measurement.

For the inclusive and differential cross section, the dependency of the correction factor on the relative contribution of each Higgs boson production mechanism is estimated by varying the relative fraction of each mode by an amount consistent with the uncertainties obtained from the combined ATLAS and CMS measurement of the Higgs boson production cross sections [10] and the ATLAS $t\bar{t}H$ measurement [14]. The correction factors are cross-checked using the alternative MADGRAPH5_AMC@NLO-FxFx ggF samples and the differences with respect to the nominal values are found to be well within the statistical uncertainties of the samples.

The dominant theoretical uncertainty in the expected ZZ^* background yield in the signal mass window is obtained by varying the factorisation and renormalisation QCD scales by factors of two. The configuration with the largest impact is chosen to define the uncertainty in each experimental category as the relative difference between the prediction in this and the nominal configuration. This uncertainty is about 4% for the inclusive event yield and is as large as 30% for the categories where additional jets are required. The impact of the QCD scale uncertainty on the BDT discriminant shapes is approximately 1–2%. The impact of the PDF uncertainty in the ZZ^* event yield in each category and on the BDT distributions, obtained using the MC replicas of the NNPDF 3.0 PDF set, was found to be approximately 1–2%. The impact of the parton shower modelling uncertainty on the ZZ^* event yield is estimated to be approximately 1–5%, with the largest value reached in the categories where the presence of one or more jets is required. In addition, the event yield and BDT discriminant shapes in each event category are compared to the data in a $m_{4\ell}$ sideband around the signal region ($m_{4\ell} < 115$ GeV or $130 \text{ GeV} < m_{4\ell} < 170$ GeV). Good agreement between the SHERPA predictions and the data is found.

For $t\bar{t} + Z$, uncertainties due to PDF and QCD scale are estimated to be around 4% and 10% respectively. An additional shape uncertainty, ranging from 3% to 25%, is applied to the $t\bar{t} + Z$ $\text{BDT}_{t\bar{t}H\text{-Had}}$ distribution to account for the difference between MADGRAPH5_AMC@NLO and SHERPA. Similar uncertainties for PDF and QCD scale are applied to the VVV background estimate.

8 Results

The observed number of events in the four decay channels after the event selection, as well as the expected signal and background yields, are presented in Table 6.

Figures 3 and 4 show the expected and observed four-lepton invariant mass distributions, inclusive and per final state. Only events with a four-lepton invariant mass in the range 115–130 GeV are used in the extraction of the signal yield. The $m_{4\ell}$ distribution shows two clear peaks corresponding to $Z \rightarrow 4\ell$

Table 6: Number of expected and observed events in the four decay channels after the event selection, in the mass range $115 \text{ GeV} < m_{4\ell} < 130 \text{ GeV}$. The sum of the expected number of SM Higgs boson events and the estimated background yields is compared to the data. Combined statistical and systematic uncertainties are included for the predictions (see Section 7).

Final state	Signal	ZZ^* background	Other backgrounds	Total expected	Observed
4μ	40.5 ± 1.7	19.0 ± 1.1	1.71 ± 0.10	61.2 ± 2.0	64
$2e2\mu$	28.2 ± 1.2	13.3 ± 0.8	1.38 ± 0.10	42.8 ± 1.4	64
$2\mu 2e$	22.1 ± 1.4	9.2 ± 0.9	2.99 ± 0.09	34.3 ± 1.7	39
$4e$	21.1 ± 1.4	8.6 ± 0.8	2.90 ± 0.09	32.5 ± 1.6	28
Total	112 ± 5	50 ± 4	8.96 ± 0.12	171 ± 6	195

production and to the Higgs boson signal with a mass near 125 GeV. The overall observed and predicted event counts agree within 1.7 standard deviations.

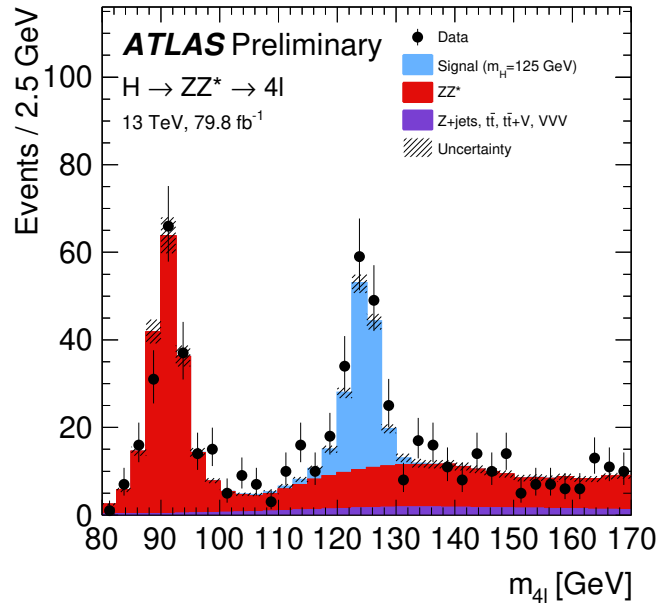


Figure 3: The expected and observed inclusive four-lepton invariant mass distributions for the selected Higgs boson candidates, shown for an integrated luminosity of 79.8 fb^{-1} and at $\sqrt{s} = 13 \text{ TeV}$. The uncertainty in the prediction is shown by the hatched band, calculated as described in Section 7.

The observed and expected distributions of the jet multiplicity and the four-lepton transverse momenta are shown in Figure 5. Further details on the compatibility with the SM are reported in Section 8.2.

The expected numbers of signal and background events in each reconstructed event category of the production mode analysis are shown in Table 7 together with the corresponding observed number of events. The expected event yields are in reasonable agreement with the observed ones. The largest differences are observed in the two VBF-enriched categories.

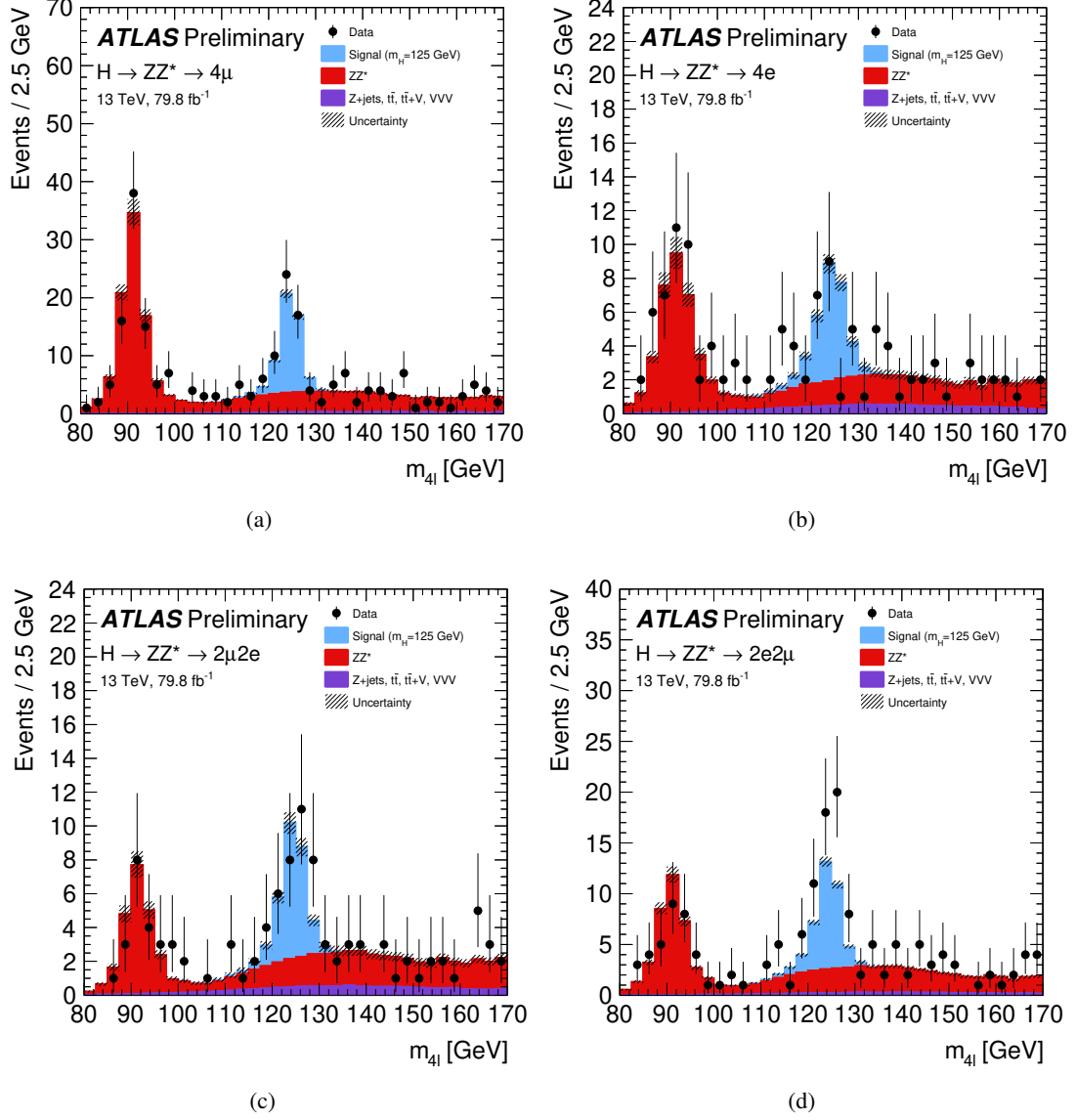


Figure 4: The expected and observed four-lepton invariant mass distribution for the selected Higgs boson candidates, for the different decay channels (a) 4μ , (b) $4e$, (c) $2\mu 2e$, (d) $2e 2\mu$. The uncertainty in the prediction is shown by the hatched band, calculated as described in Section 7.

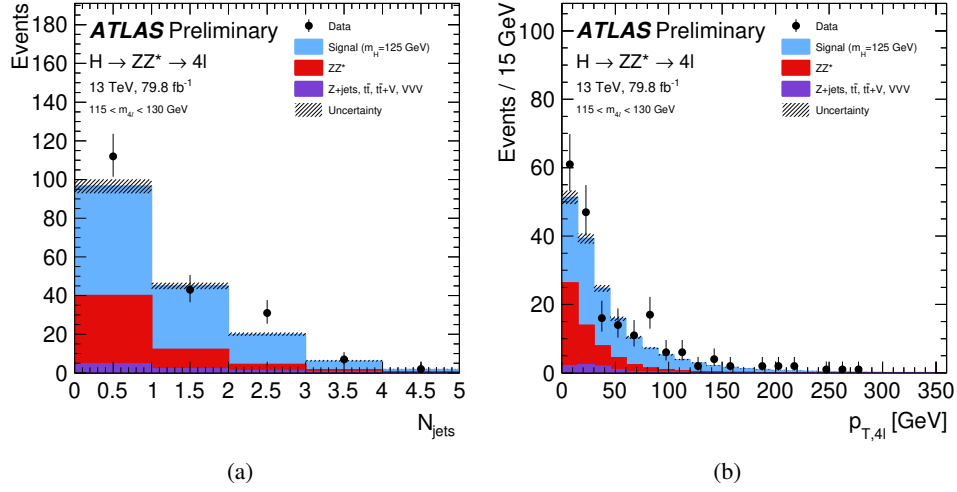


Figure 5: The observed and expected distributions of (a) N_{jets} after the inclusive selection, (b) $p_{T,4\ell}$ for an integrated luminosity of 79.8 fb^{-1} collected at $\sqrt{s} = 13 \text{ TeV}$ assuming the SM Higgs boson signal with a mass $m_H = 125 \text{ GeV}$. The uncertainty in the prediction is shown by the hatched band, calculated as described in Section 7.

Table 7: The expected and observed numbers of signal and background events in the mass range $115 < m_{4\ell} < 130 \text{ GeV}$ for an integrated luminosity of 79.8 fb^{-1} and at $\sqrt{s} = 13 \text{ TeV}$ in each reconstructed event category assuming the SM Higgs boson signal with a mass $m_H = 125 \text{ GeV}$. The sum of the expected number of SM Higgs boson events and the estimated background yields is compared to the data. Combined statistical and systematic uncertainties are included for the predictions. Compared to Ref. [114], updated jet uncertainties with smaller impact are used.

Reconstructed event category	Signal	ZZ^* background	Other backgrounds	Total expected	Observed
$0j\text{-}p_T^{4\ell}\text{-Low}$	56 ± 5	35.2 ± 2.5	4.1 ± 0.4	95 ± 6	112
$1j\text{-}p_T^{4\ell}\text{-Low}$	17.9 ± 2.2	7.6 ± 1.1	1.36 ± 0.14	26.9 ± 2.5	25
$1j\text{-}p_T^{4\ell}\text{-Med}$	11.5 ± 1.5	2.25 ± 0.32	0.72 ± 0.08	14.5 ± 1.5	14
$1j\text{-}p_T^{4\ell}\text{-High}$	3.3 ± 0.5	0.31 ± 0.05	0.198 ± 0.028	3.8 ± 0.5	4
$\text{VBF-enriched-}p_T^j\text{-Low}$	12.8 ± 1.6	2.8 ± 0.7	1.04 ± 0.10	16.6 ± 1.8	24
$\text{VBF-enriched-}p_T^j\text{-High}$	1.25 ± 0.20	0.28 ± 0.09	0.155 ± 0.025	1.68 ± 0.23	4
VH-Had-enriched	7.2 ± 1.1	1.6 ± 0.4	0.59 ± 0.07	9.4 ± 1.2	11
VH-Lep-enriched	0.70 ± 0.05	0.068 ± 0.013	0.035 ± 0.008	0.80 ± 0.05	1
$0j\text{-}p_T^{4\ell}\text{-High}$	0.183 ± 0.025	0.0082 ± 0.0027	0.164 ± 0.023	0.355 ± 0.034	0
$t\bar{t}H\text{-Had-enriched}$	0.60 ± 0.05	0.035 ± 0.016	0.194 ± 0.030	0.83 ± 0.06	0
$t\bar{t}H\text{-Lep-enriched}$	0.238 ± 0.026	0.0005 ± 0.0005	0.0144 ± 0.0034	0.253 ± 0.026	0
Total	112 ± 5	50 ± 4	8.96 ± 0.12	171 ± 6	195

The expected and observed distributions of the BDT discriminants are shown in Figure 6, where a small excess is observed at larger values of the BDT_{VBF} . No events are observed in the $0j\text{-}p_{\text{T}}^{4\ell}\text{-High}$, $t\bar{t}H\text{-Had}$ -enriched and $t\bar{t}H\text{-Lep}$ -enriched categories. All distributions are in good agreement with the data. The fit results and compatibility with the SM are discussed in Section 8.3. The event displays of the selected candidates falling in the $VH\text{-Lep}$ category and in $VH\text{-Had}$ with the highest BDT score are shown in Figures 7, 8 and 9.

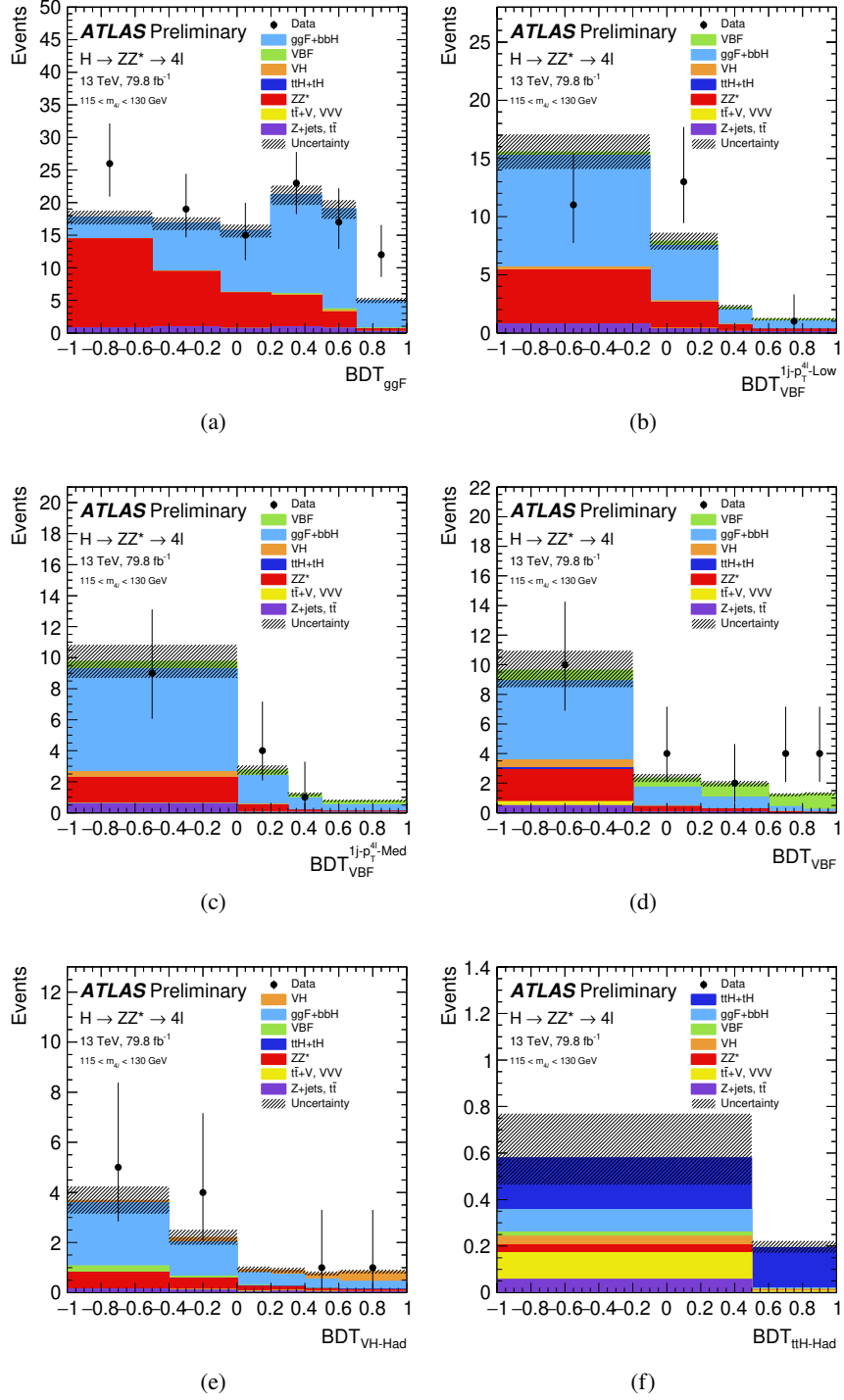


Figure 6: The observed and expected BDT output distributions in the (a) $0j-p_T^{4\ell}$ -Low, (b) $1j-p_T^{4\ell}$ -Low, (c) $1j-p_T^{4\ell}$ -Med, (d) VBF-enriched- p_T^j -Low, (e) VH -Had-enriched and (f) ttH -Had-enriched categories for an integrated luminosity of 79.8 fb^{-1} and at $\sqrt{s} = 13$ TeV assuming the SM Higgs boson signal with a mass $m_H = 125$ GeV. No events are observed in the ttH -Had-enriched category, while a small excess is observed at larger values of BDT_{VBF} . The bin boundaries are chosen to maximise the significance of the targeted signal in each category. The uncertainty in the prediction is shown by the hatched band, calculated as described in Section 7.

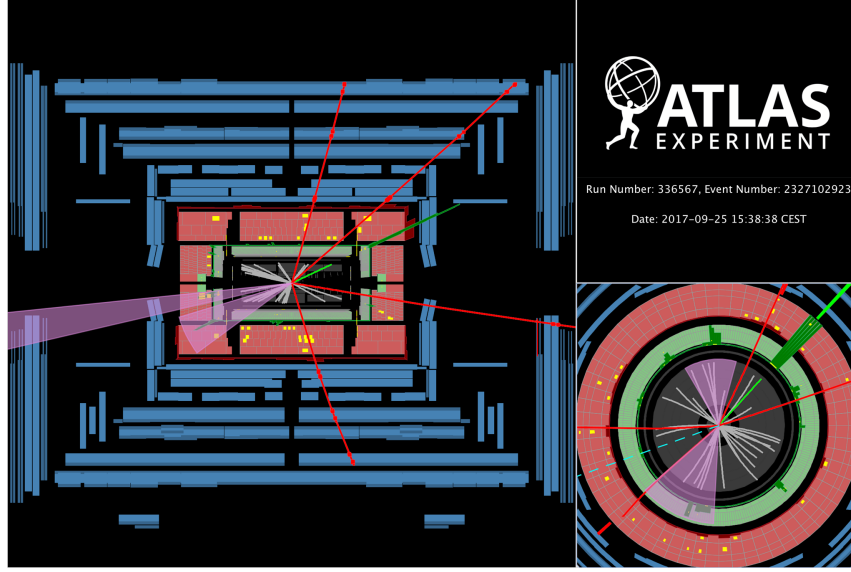


Figure 7: Event display of the 4μ VH -Lep candidate. The invariant mass of the 4-lepton system is 124.6 GeV, the muons are indicated by 4 red tracks ($p_T = 45.7, 25.3, 15.4$ and 9.7 GeV), while the extra electron by a green track ($p_T = 78.9$ GeV), the cyan dashed line represents the direction of the E_T^{miss} vector ($E_T^{\text{miss}} = 49.4$ GeV), and there are two jets marked with purple cones ($p_{T,j1} = 76.1$ GeV and $p_{T,j2} = 41.5$ GeV).

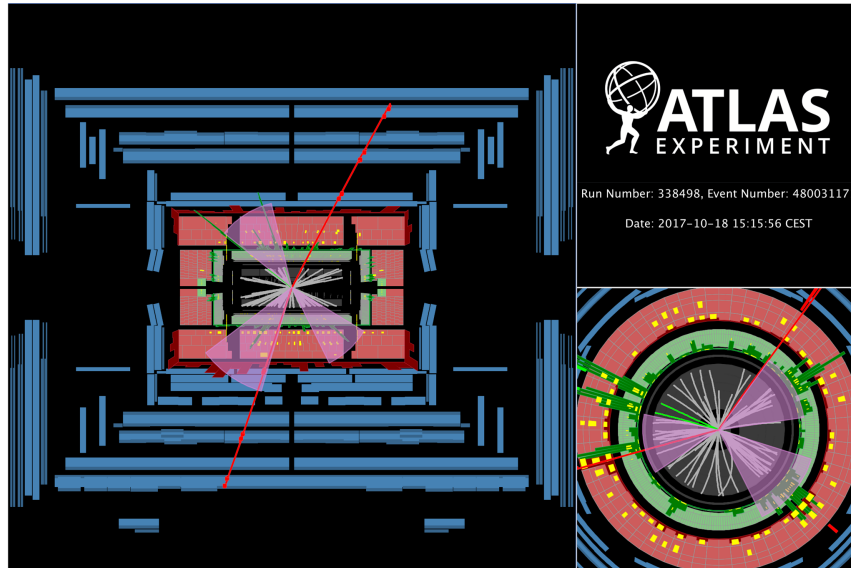


Figure 8: Event display of the $2\mu 2e$ VH -Had candidate with $\text{BDT}_{VH\text{-Had}} = 0.70$. The invariant mass of the 4-lepton system is 123.3 GeV, the muons ($p_T = 62.2$ and 30.3 GeV) are indicated by two red tracks while the two electrons by a green track ($p_T = 22.9$ and 15.4 GeV). The three jets, with an invariant mass of two leading jets equal to 89.0 GeV, are marked with purple cones ($p_{T,j1} = 85.0$ GeV, $p_{T,j2} = 72.9$ GeV and $p_{T,j3} = 54.1$ GeV).

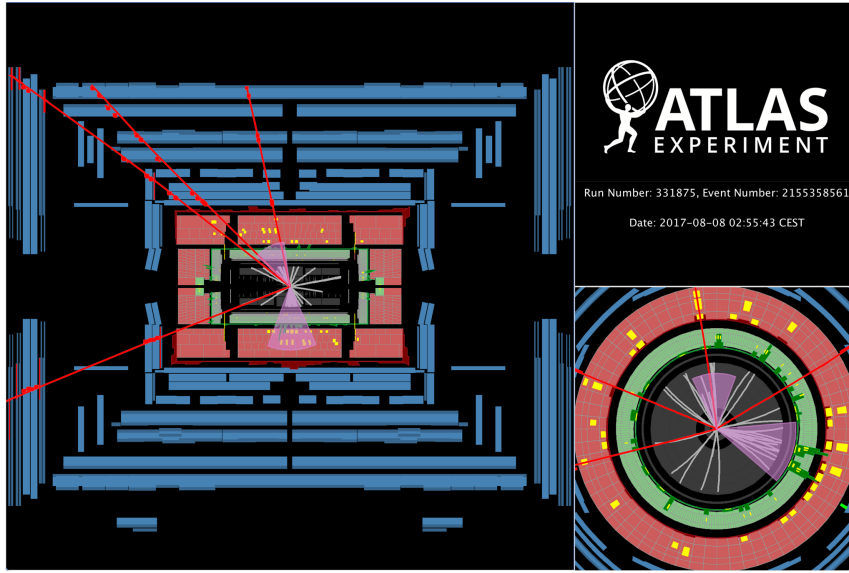


Figure 9: Event display of the 4μ VH -Had candidate with the with $\text{BDT}_{VH\text{-}Had} = 0.47$. The invariant mass of the 4-lepton system is 128.2 GeV, the muons are indicated by 4 red tracks ($p_T = 103.7, 16.9, 16.4$ and 15.4 GeV). The two jets, with an invariant mass of 96.2 GeV, are marked with purple cones ($p_{T,j1} = 64.9$ GeV and $p_{T,j2} = 37.9$ GeV).

8.1 Statistical analysis

The inclusive fiducial, differential and productions cross sections are measured using a binned profile-likelihood-ratio fit [126], taking into account all bins of a given distribution in all categories. The likelihood function includes the shape and normalisation uncertainties, as outlined in Section 7, of the signal and backgrounds predictions as nuisance parameters. The cross sections are extracted by minimising twice the negative logarithm of the profile likelihood ratio, $-2 \ln \Lambda$. In the asymptotic assumption, i.e. the large sample limit, $-2 \ln \Lambda$ behaves as a χ^2 distribution with one degree of freedom. The compatibility of a measured cross section and a theoretical prediction is evaluated by computing a p -value based on the difference between the value of $-2 \ln \Lambda$ at the best-fit value and the value obtained by fixing the cross sections in all bins to the ones predicted by the theory. These p -values do not include the uncertainties in the theoretical predictions, which are significantly smaller than the total data uncertainties. For all measured observables the asymptotic assumption is verified with pseudo-experiments, and where the number of observed events is less than 3, the uncertainties are corrected to the values obtained with the pseudo-experiments.

For the inclusive and differential cross section, the $m_{4\ell}$ distribution is fitted in each channel and bin and the cross section is measured from the fit results following Eq. (1). The fiducial cross sections of the four final states can either be summed together to obtain an inclusive fiducial cross section, or they can be combined assuming the SM $ZZ^* \rightarrow 4\ell$ relative branching ratios. The latter combination is more model dependent, but benefits from a smaller statistical uncertainty.

For the production cross section, the BDT distributions and observed yields are fitted in the given reconstructed event categories. The cross sections are treated as independent parameters of interest for each production bin and are correlated among the different event categories. The cross section parameter of interest in each production bin is alternatively replaced by $\mu \cdot \sigma_{\text{SM}}$, allowing for an interpretation in terms of the signal strength μ relative to the SM prediction σ_{SM} .

8.2 Inclusive and differential cross section measurement

The inclusive fiducial production cross sections of $H \rightarrow ZZ^* \rightarrow 4\ell$ are presented in Table 8 and Figure 10. The left panel in Figure 10 shows the fiducial cross sections for the four individual decay channels for 4μ , $4e$ decays (hereafter defined as same flavour), and for $2\mu 2e$, $2e 2\mu$ decays (hereafter defined as opposite flavour). The middle panel shows the cross sections for opposite- and same-flavour decays, which can provide a handle on same-flavour interference effects, as well as the fiducial cross sections obtained by either summing all 4ℓ decay channels or combining them assuming SM relative branching ratios. The data are compared to the SM prediction after accounting for the fiducial acceptance as determined from the SM Higgs boson simulated samples (see Section 3). The fiducial cross section is extrapolated to the total phase space, as shown in the right panel of Figure 10, using the same fiducial acceptance as well as the branching ratios, with the uncertainties described in Section 7. The total cross section is also compared to the cross sections predicted by NNLOPS, MADGRAPH5_AMC@NLO-FxFx and HRES 2.3 [53, 127] for ggF, while for all other productions modes the predictions described in Section 3 are used. For ggF, all generators predict cross sections that are lower than the N³LO calculation. The p -values, calculated as described in Section 6.1.2, are shown in Table 8.

The measured differential production cross sections and their comparisons to SM predictions are presented in Figure 11. The data are compared to SM expectations constructed from the ggF predictions provided

Table 8: The fiducial and total cross sections of Higgs boson production measured in the 4ℓ final state. The fiducial cross sections are given separately for each decay channel, and for same- and opposite-flavour decays. The inclusive fiducial cross section is measured as the sum of all channels (σ_{sum}), as well as by combining the per-channel measurements assuming SM $ZZ^* \rightarrow 4\ell$ relative branching ratios (σ_{comb}). For the total cross section (σ_{tot}), the Higgs boson branching ratio at 125 GeV is assumed. The total SM prediction is accurate to N³LO in QCD for the ggF process. For the fiducial cross section predictions, the SM cross sections are multiplied by the acceptances determined using the NNLOPS sample for ggF. For all the other production modes, the predictions from the samples discussed in Section 3 are added. The p -values indicating the compatibility of the measurement and the SM prediction are shown as well. They do not include the systematic uncertainty in the theoretical predictions.

Cross section [fb]	Data ($\pm$ (stat.) $\pm$ (syst.))	Standard Model prediction	p -value [%]
$\sigma_{4\mu}$	0.97 $\pm$ 0.17 $\pm$ 0.05	0.886 ± 0.039	62
σ_{4e}	0.61 $\pm$ 0.21 $\pm$ 0.07	0.886 ± 0.039	25
$\sigma_{2\mu 2e}$	0.88 $\pm$ 0.21 $\pm$ 0.08	0.786 ± 0.035	66
$\sigma_{2e 2\mu}$	1.37 $\pm$ 0.22 $\pm$ 0.07	0.786 ± 0.035	0.3
$\sigma_{4\mu+4e}$	1.58 $\pm$ 0.27 $\pm$ 0.10	1.77 ± 0.07	51
$\sigma_{2\mu 2e+2e 2\mu}$	2.26 $\pm$ 0.31 $\pm$ 0.13	1.57 ± 0.06	2.4
σ_{sum}	3.84 $\pm$ 0.41 $\pm$ 0.23	3.35 ± 0.15	27
σ_{comb}	4.04 $\pm$ 0.41 $\pm$ 0.22	3.35 ± 0.15	12
σ_{tot} [pb]	67.2 $\pm$ 6.8 $\pm$ 4.1	55.7 ± 2.5	13

by NNLOPS and MADGRAPH5_AMC@NLO-FxFx. All samples are normalised to the most accurate SM predictions, as discussed in Section 3. The shaded bands on the expected cross sections indicate the PDF and scale uncertainties. The figures include the p -values quantifying the compatibility of the measurement and the SM predictions.

8.3 Production mode cross sections

Assuming that the relative signal fractions in each production bin are given by the predictions for the SM Higgs boson, the inclusive $H \rightarrow ZZ^*$ production cross section for $|y_H| < 2.5$ is measured to be:

$$\sigma \cdot \mathcal{B} \equiv \sigma \cdot \mathcal{B}(H \rightarrow ZZ^*) = 1.57 \pm 0.15(\text{stat.}) \pm 0.08(\text{exp.}) \pm 0.04(\text{th.}) \text{ pb} = 1.57 \pm 0.18 \text{ pb}.$$

The SM prediction is $(\sigma \cdot \mathcal{B})_{\text{SM}} \equiv (\sigma \cdot \mathcal{B}(H \rightarrow ZZ^*))_{\text{SM}} = 1.33 \pm 0.09 \text{ pb}$. The data are also interpreted in terms of the global signal strength, yielding

$$\mu = 1.19 \pm 0.12(\text{stat.}) \pm 0.06(\text{exp.})^{+0.08}_{-0.07}(\text{th.}) = 1.19^{+0.16}_{-0.15}.$$

The measured cross section and signal strength agree with the SM prediction at the level of 1.5 and 1.4 standard deviations, respectively. The corresponding likelihood functions are shown in Figure 12. The dominant systematic uncertainty in the cross section measurement is the experimental uncertainty in the lepton efficiency and integrated luminosity measurements. The signal strength measurement is also equally affected by the theoretical uncertainty of the ggF signal yield due to QCD scale variations.

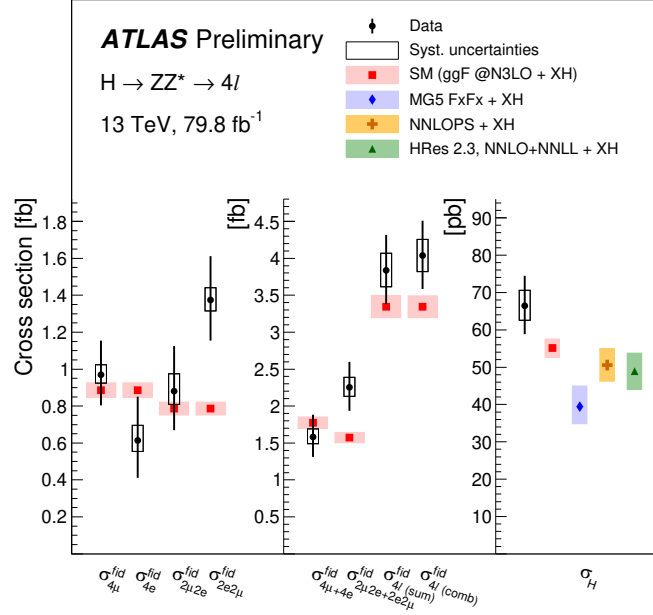


Figure 10: The fiducial cross sections (left two panels) and total cross section (right panel) of Higgs boson production measured in the 4ℓ final state. The fiducial cross sections are shown separately for each decay channel, and for same- and opposite-flavour decays. The inclusive fiducial cross section is measured as the sum of all channels, as well as by combining the per-channel measurements assuming SM $ZZ^* \rightarrow 4\ell$ relative branching ratios. The total SM prediction is accurate to N³LO in QCD for the ggF process. The cross sections for all other Higgs boson production modes XH are added. For the fiducial cross section predictions, the SM cross sections are multiplied by the acceptances determined using the NNLOPS sample for ggF and the samples discussed in Section 3 for the other production modes. For the total cross section, the predictions by the generators NNLOPS, HRES, and MADGRAPH5_AMC@NLO-FxFx are also shown. The error bars on the data points show the total uncertainties, while the systematic uncertainties are indicated by the boxes. The shaded bands around the theoretical predictions indicate the PDF and scale uncertainties, calculated as described in Section 7.2.

The SM expected cross section, the observed values of $\sigma \cdot \mathcal{B}(H \rightarrow ZZ^*)$ and their ratio for the inclusive production and in each Stage-0 and reduced Stage-1 production bin are shown in Table 9. The corresponding values are summarised in Figure 13. The bbH production process is treated as a part of the ggF production bins. In the ratio calculation, uncertainties on the SM expectation are not taken into account.

All Stage-0 and reduced Stage-1 ggF measurements agree with the predictions for the SM Higgs boson within 2 standard deviation. Due to the limited number of events in the $t\bar{t}H$ -enriched categories, only upper limits are set on the cross section and signal strengths for the $t\bar{t}H$ production. The limits are based on the CL_s prescription [128] and derived using pseudo-experiments. The $t\bar{t}H$ parameter of interest is constrained to positive values to avoid the fit's prediction of negative total event yields in the $t\bar{t}H$ -enriched categories and provide a stable fit configuration. The impact of this constraint on the final fit results is negligible.

The dominant contribution to the measurement uncertainty in the Stage-0 ggF production bin originates from the same sources as in the inclusive measurement. In the remaining three Stage-0 production bins, similar sources of uncertainty are relevant for both the cross section and the signal strength measurements:

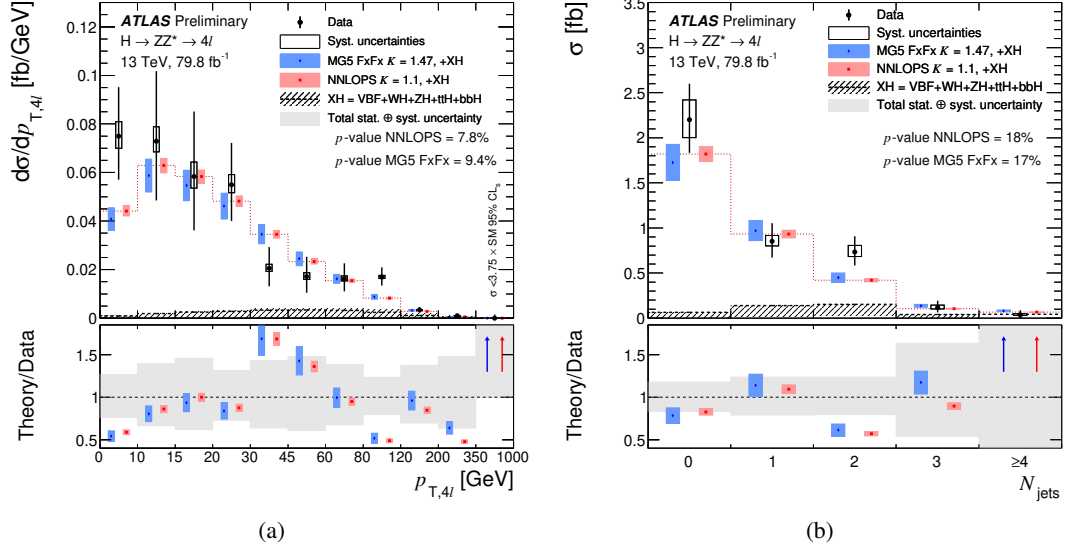


Figure 11: Differential fiducial cross sections, for (a) the transverse momentum $p_{T,4\ell}$ of the Higgs boson, (b) the number of jets N_{jets} . The measured cross sections are compared to ggF predictions by NNLOPS and MADGRAPH5_AMC@NLO-FxFx normalised to the N³LO total cross section with the listed K -factors. MC based predictions for all other Higgs boson production modes XH are normalised to the SM predictions. The error bars on the data points show the total uncertainties, while the systematic uncertainties are indicated by the boxes. The shaded bands on the expected cross sections indicate the PDF and scale uncertainties, calculated as described in Section 7.2. The p -values indicating the compatibility of the measurement and the SM prediction are shown as well. They do not include the systematic uncertainty in the theoretical predictions. The bottom part of the figures shows the ratio of different predictions to the data, the grey area represents the total uncertainty of the measurement.

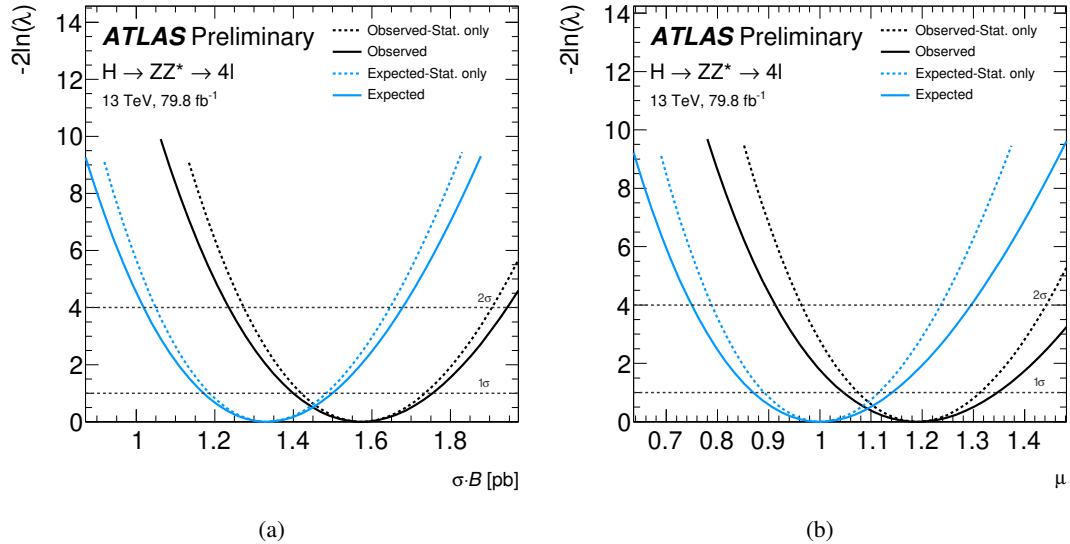
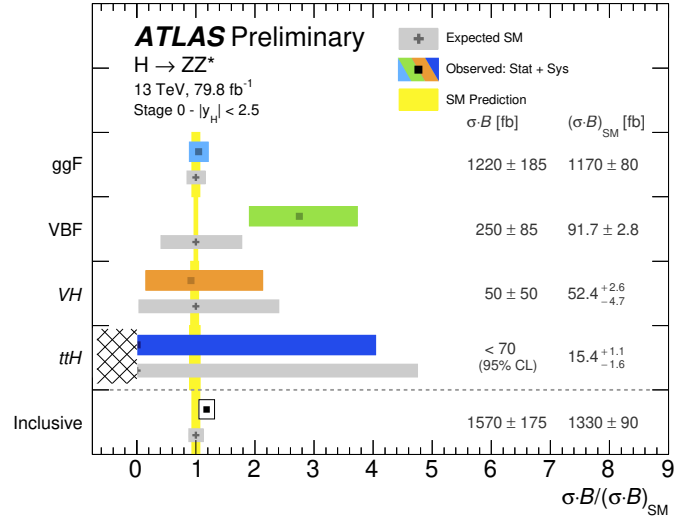


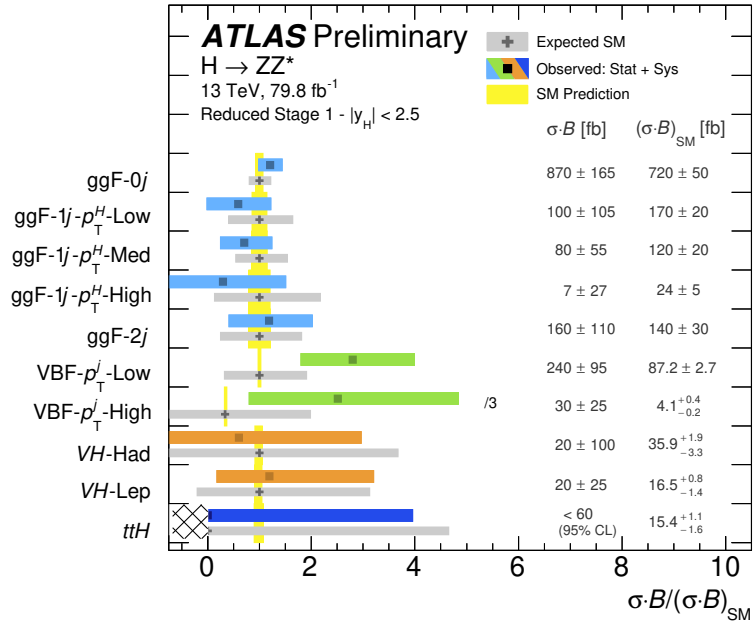
Figure 12: The expected (blue) and observed (black) profile likelihood as a function of (a) $\sigma \cdot \mathcal{B}(H \rightarrow ZZ^*)$ and (b) the inclusive signal strength μ ; the scans are shown both with (solid line) and without (dashed line) systematic uncertainties.

Table 9: The SM expected cross section $(\sigma \cdot \mathcal{B})_{\text{SM}}$, the observed values of $\sigma \cdot \mathcal{B}$, and their ratio $(\sigma \cdot \mathcal{B})/(\sigma \cdot \mathcal{B})_{\text{SM}}$ for the inclusive production and for each Stage-0 and reduced Stage-1 production bin for the $H \rightarrow ZZ^*$ decay for an integrated luminosity of 79.8 fb^{-1} and at $\sqrt{s} = 13 \text{ TeV}$. The bbH (tH) contribution is included in the ggF (ttH) production bins. The upper limits correspond to the 95% CL obtained with pseudo-experiments using the CL_s method. The uncertainties are given as (stat.)+(exp.)+(th.) for the inclusive cross section and Stage 0 and as (stat.)+(syst.) for reduced Stage 1. Stage 1 results are dominated by statistical error and the impact of theory uncertainties is smaller compared to the Stage 0. The actual impact of the theory uncertainties for Stage 1 are smaller than the rounding.

Production bin	Cross section $(\sigma \cdot \mathcal{B})$ [pb]		$(\sigma \cdot \mathcal{B})/(\sigma \cdot \mathcal{B})_{\text{SM}}$
	SM expected	Observed	Observed
Inclusive production, $ y_H < 2.5$			
	1.33 ± 0.09	$1.57 \pm 0.16 \pm 0.07 \pm 0.04$	$1.18 \pm 0.12 \pm 0.05 \pm 0.03$
Stage-0 production bins, $ y_H < 2.5$			
ggF	1.17 ± 0.08	$1.22 \pm 0.17 \pm 0.07 \pm 0.04$	$1.04 \pm 0.14 \pm 0.06 \pm 0.03$
VBF	0.0917 ± 0.0028	$0.25 \pm 0.08 \pm 0.02 \pm 0.01$	$2.8 \pm 0.9 \pm 0.2 \pm 0.2$
VH	$0.0524^{+0.0026}_{-0.0047}$	$0.05 \pm 0.05 \pm 0.01 \pm 0.01$	$0.9 \pm 1.0 \pm 0.1 \pm 0.1$
ttH	$0.0154^{+0.0011}_{-0.0016}$	< 0.07	< 4.04
Reduced Stage-1 production bins, $ y_H < 2.5$			
ggF-0j	0.72 ± 0.05	$0.87 \pm 0.14 \pm 0.08$	$1.20 \pm 0.20 \pm 0.10$
ggF-1j- p_T^H -Low	0.170 ± 0.020	$0.10 \pm 0.10 \pm 0.05$	$0.59 \pm 0.55 \pm 0.27$
ggF-1j- p_T^H -Med	0.120 ± 0.020	$0.08 \pm 0.06 \pm 0.02$	$0.70 \pm 0.48 \pm 0.13$
ggF-1j- p_T^H -High	0.024 ± 0.005	$0.007 \pm 0.026 \pm 0.006$	$0.3 \pm 1.1 \pm 0.3$
ggF-2j	0.140 ± 0.030	$0.16 \pm 0.10 \pm 0.03$	$1.19 \pm 0.76 \pm 0.26$
VBF- p_T^j -Low	0.0872 ± 0.0027	$0.24 \pm 0.09 \pm 0.02$	$2.8 \pm 1.0 \pm 0.2$
VBF- p_T^j -High	$0.0041^{+0.0004}_{-0.0002}$	$0.03 \pm 0.02 \pm 0.01$	$7.5 \pm 6.0 \pm 0.7$
VH-Had	$0.0359^{+0.0019}_{-0.0033}$	$0.02 \pm 0.10 \pm 0.01$	$0.6 \pm 2.9 \pm 0.3$
VH-Lep	$0.0165^{+0.0008}_{-0.0014}$	$0.02 \pm 0.03 \pm 0.01$	$1.2 \pm 1.5 \pm 0.1$
ttH	$0.0154^{+0.0011}_{-0.0016}$	< 0.06	< 4.02



(a)



(b)

Figure 13: The observed and expected SM values of the cross section ratios $\sigma \cdot \mathcal{B}$ normalised by the SM expectation $(\sigma \cdot \mathcal{B})_{\text{SM}}$ for the inclusive production and in the (a) Stage-0 and (b) reduced Stage-1 production bins for an integrated luminosity of 79.8 fb⁻¹ at $\sqrt{s} = 13$ TeV. Different colours for the observed results indicate different Higgs boson production modes. The hatched area indicates that the ttH parameter of interest is constrained to positive values. For visualisation purposes, the VBF- p_T^j -High value is divided by a factor of three when shown normalised to $(\sigma \cdot \mathcal{B})_{\text{SM}}$. The yellow vertical band represents the theory uncertainty in the signal prediction, while the horizontal grey bands represent the expected measurement uncertainty.

the jet energy scale or resolution uncertainties in all production bins and, additionally, theoretical jet bin migration in the ggF process for the VBF and VH production bins. For the reduced Stage-1 categories the dominant cross section uncertainties are the integrated luminosity and lepton efficiency measurements for the VH -Lep bin and the jet energy scale or resolution for all other categories. The ggF- $1j$ - p_T^H -Med, ggF- $1j$ - p_T^H -High, VBF- p_T^j -Low and ttH bins are additionally affected by parton shower uncertainties, while the uncertainties related to the top-quark mass treatment have a dominant impact on the VBF- p_T^j -High bin. The signal strength measurement of the reduced Stage-1 ggF processes is also strongly affected by the theory uncertainties from event migrations between different jet multiplicity and Higgs boson transverse momentum bins, while parton shower and QCD scale uncertainties affect the remaining reduced Stage-1 production bins.

Figure 14 shows the likelihood contours in the $(\sigma_{\text{ggF}} \cdot \mathcal{B}(H \rightarrow ZZ^*), \sigma_{\text{VBF}} \cdot \mathcal{B}(H \rightarrow ZZ^*))$ and $(\sigma_{\text{VBF}} \cdot \mathcal{B}(H \rightarrow ZZ^*), \sigma_{\text{VH}} \cdot \mathcal{B}(H \rightarrow ZZ^*))$ plane. The other cross section parameters are left free in the fit, i.e. they are not treated as parameters of interest. The compatibility with respect to the Standard Model expectation is at the level of 2.1 and 1.9 standard deviations in the $(\sigma_{\text{ggF}} \cdot \mathcal{B}(H \rightarrow ZZ^*), \sigma_{\text{VBF}} \cdot \mathcal{B}(H \rightarrow ZZ^*))$ and $(\sigma_{\text{VBF}} \cdot \mathcal{B}(H \rightarrow ZZ^*), \sigma_{\text{VH}} \cdot \mathcal{B}(H \rightarrow ZZ^*))$ plane respectively, due to the discrepancies observed in the VBF-related distributions in Figure 6.

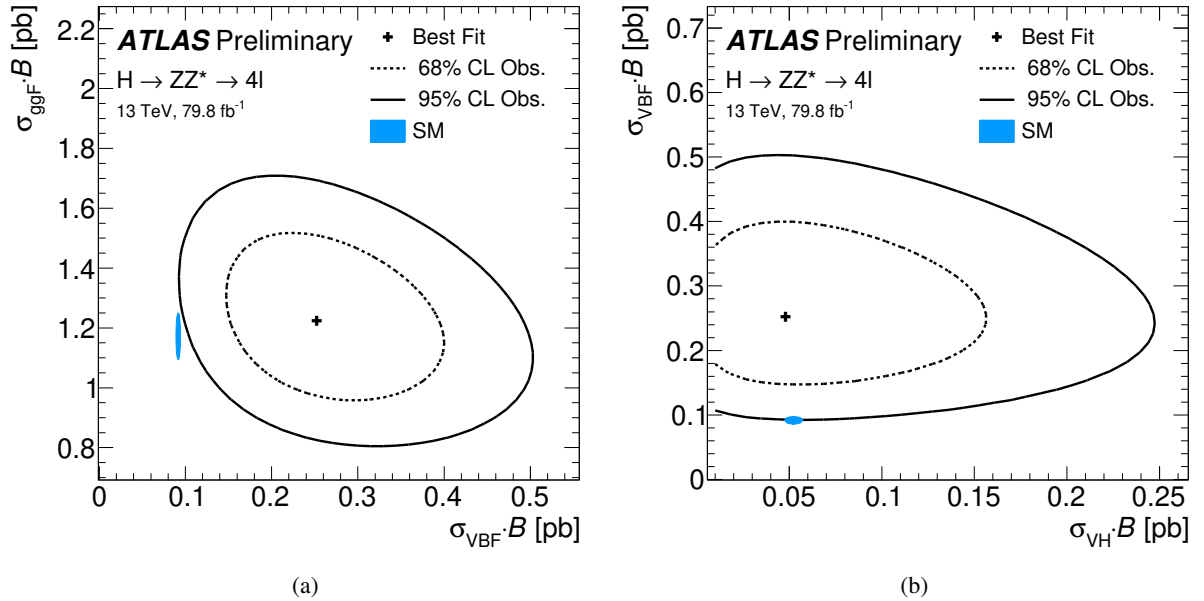


Figure 14: Likelihood contours at 68% CL (dashed line) and 95% CL (solid line) in the (a) $(\sigma_{\text{ggF}} \cdot \mathcal{B}(H \rightarrow ZZ^*), \sigma_{\text{VBF}} \cdot \mathcal{B}(H \rightarrow ZZ^*))$ and (b) $(\sigma_{\text{VBF}} \cdot \mathcal{B}(H \rightarrow ZZ^*), \sigma_{\text{VH}} \cdot \mathcal{B}(H \rightarrow ZZ^*))$ plane. The SM prediction is shown together with its theory uncertainty (filled blue ellipse).

9 Conclusion

Measurements of the inclusive fiducial, differential and production mode cross section of Higgs boson production in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel are presented. They are based on 79.8 fb $^{-1}$ of $\sqrt{s} = 13$ TeV

proton–proton collisions recorded by the ATLAS detector at the LHC in 2015–2017.

The inclusive fiducial cross section in the $H \rightarrow ZZ^* \rightarrow 4\ell$ decay channel is measured to be $\sigma_{\text{fid}} = 4.04 \pm 0.41(\text{stat.}) \pm 0.22(\text{syst.})$ fb, in agreement with the Standard Model prediction $\sigma_{\text{fid,SM}} = 3.35 \pm 0.15$ fb. The inclusive fiducial cross section is also extrapolated to the total phase space. The transverse momentum $p_{T,4\ell}$ and the number of jets N_{jets} differential fiducial production cross sections are measured. Good agreement is found between the data and the predictions of the Standard Model.

Furthermore the Higgs boson candidate events are classified into several categories according to their topologies, providing sensitivity to different production modes in various regions of phase space. Additional BDT discriminants are used to further improve the sensitivity in reconstructed event categories. The production cross sections times branching ratio for $H \rightarrow ZZ^*$ decay measured in dedicated production bins are in agreement with the SM predictions. The inclusive cross section in the Higgs boson rapidity range of $|y_H| < 2.5$ is measured to be $\sigma \cdot \mathcal{B}(H \rightarrow ZZ^*) = 1.57 \pm 0.18$ pb in agreement with the SM prediction of 1.33 ± 0.09 pb.

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Appendix

Figures 15-18 show the expected and observed distributions for the invariant mass of the lepton pairs m_{Z1} and m_{Z2} , the number of jets N_{jets} and the transverse momentum of the 4-lepton system p_T , the rapidity of the 4-lepton system y and the invariant mass of the two leading jets m_{jj} , and the BDT distribution of the $0j\text{-}p_T^{4\ell}$ -Low category and the BDT distribution of the $1j\text{-}p_T^{4\ell}$ -Low category. These distributions are obtained in a non-resonant ZZ^* -enriched control region, which is obtained by applying the full event selection except for the $m_{4\ell}$ window, i.e. $m_{4\ell} < 115$ GeV or $130 < m_{4\ell} < 170$ GeV. The expected and observed yields are presented in Table 10.

Figure 19 and Figure 20 show the expected and observed correlation and covariances matrices.

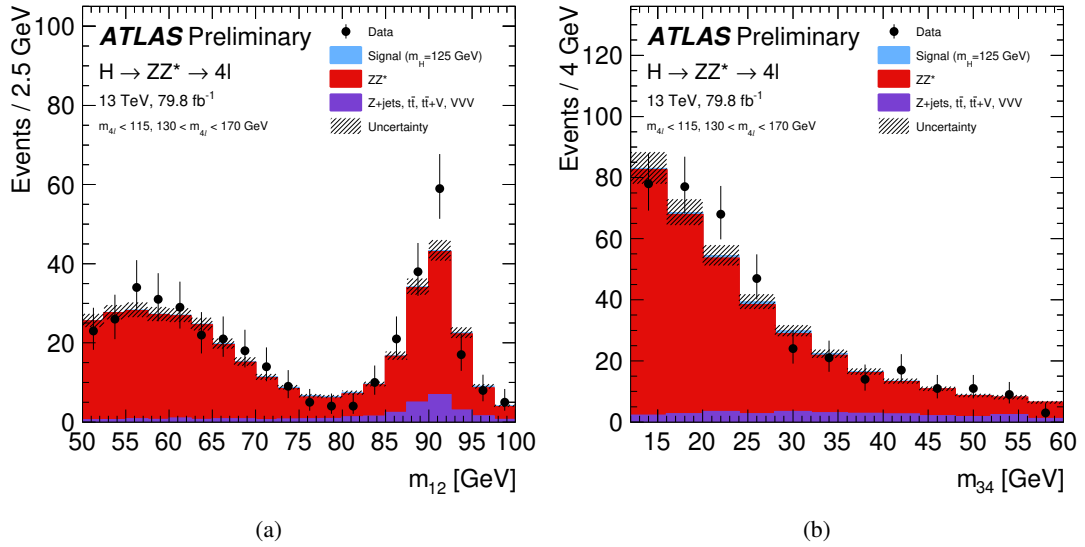


Figure 15: Reconstructed event yields of the invariant mass of the lepton pairs (a) m_{Z1} and (b) m_{Z2} , in a non-resonant ZZ^* -enriched control region, obtained by applying the full event selection except for the $m_{4\ell}$ window, i.e. $m_{4\ell} < 115$ GeV or $130 < m_{4\ell} < 170$ GeV. The error bars on the data points indicate the statistical uncertainty. The uncertainty in the prediction is shown by the dashed band.

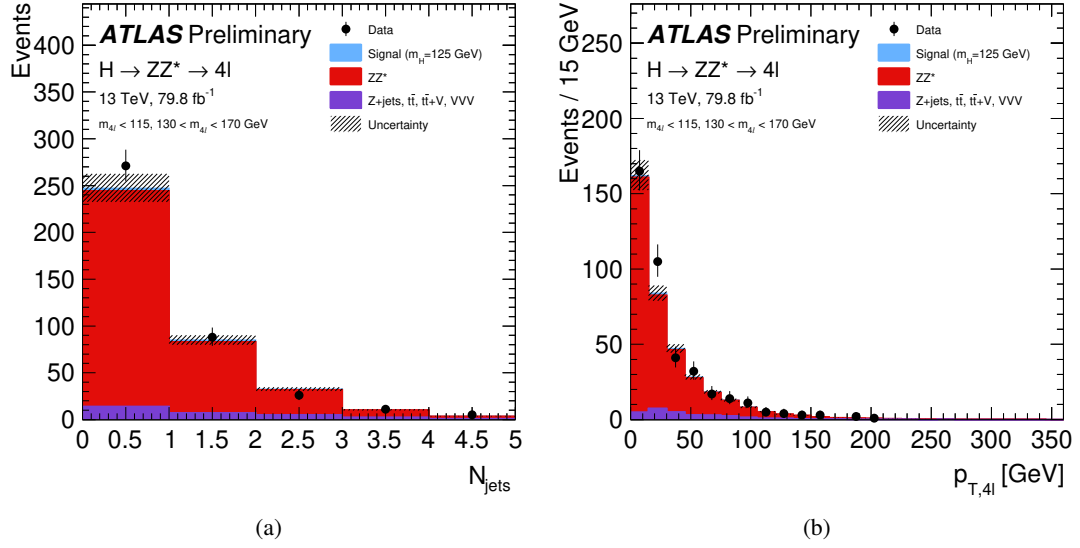


Figure 16: Reconstructed event yields of (a) the number of jets N_{jets} and (b) the transverse momentum of the 4-lepton system p_T , in a non-resonant ZZ^* -enriched control region, obtained by applying the full event selection except for the $m_{4\ell}$ window, i.e. $m_{4\ell} < 115$ GeV or $130 < m_{4\ell} < 170$ GeV. The error bars on the data points indicate the statistical uncertainty. The uncertainty in the prediction is shown by the dashed band.

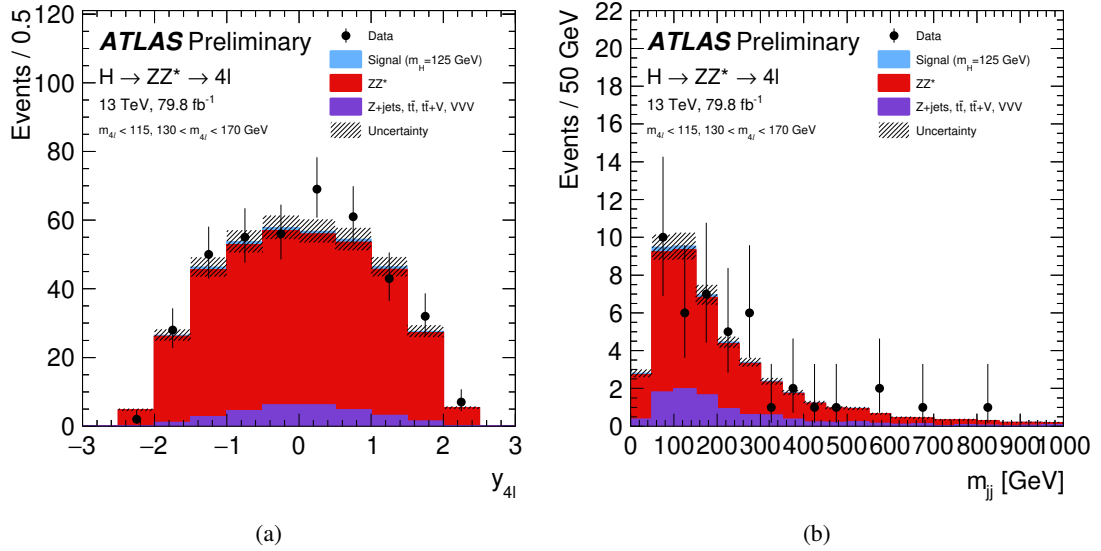


Figure 17: Reconstructed event yields of (a) the rapidity of the 4-lepton system y and (b) the invariant mass of the two leading jets m_{jj} , in a non-resonant ZZ^* -enriched control region, obtained by applying the full event selection except for the $m_{4\ell}$ window, i.e. $m_{4\ell} < 115$ GeV or $130 < m_{4\ell} < 170$ GeV. The error bars on the data points indicate the statistical uncertainty. The uncertainty in the prediction is shown by the dashed band.

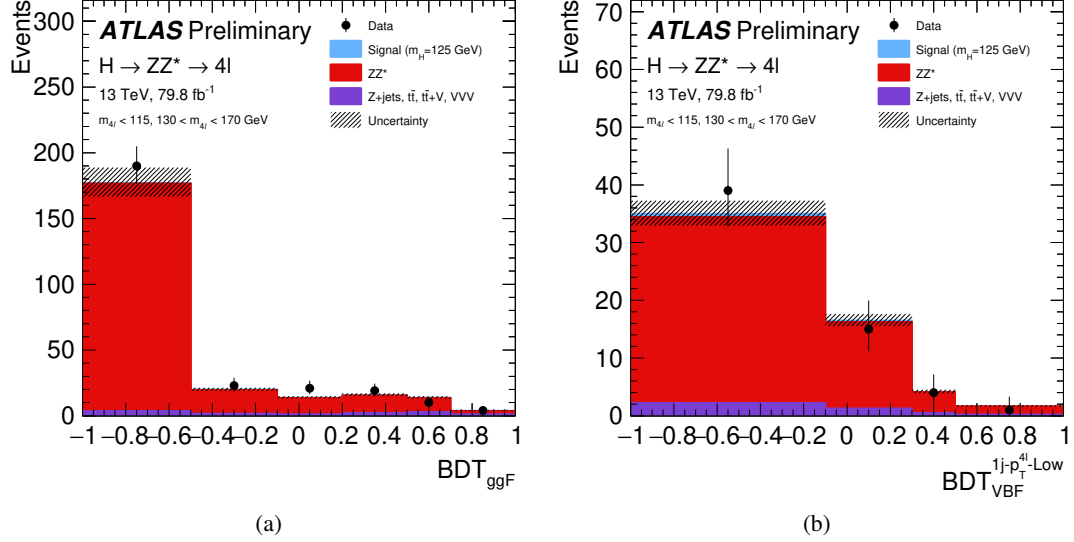
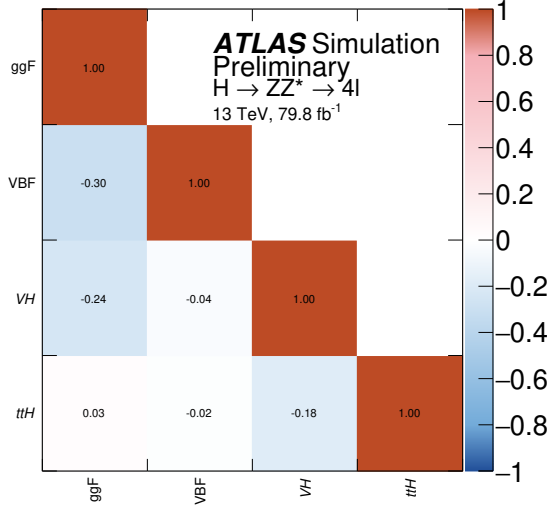


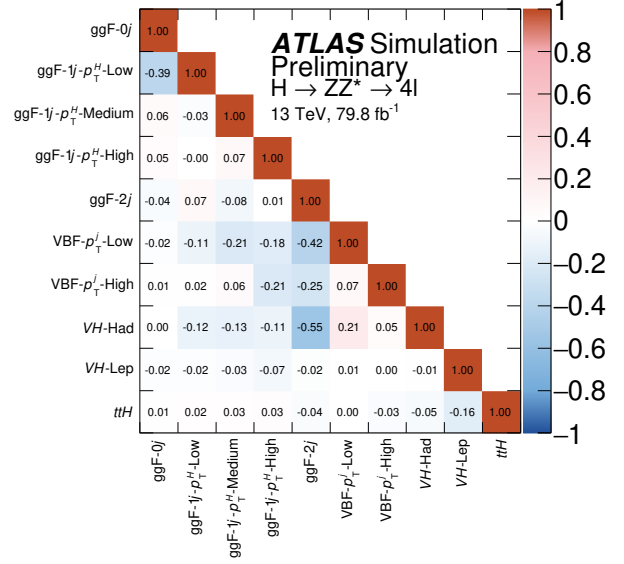
Figure 18: Reconstructed event yields of (a) the BDT distribution of the $0j\text{-}p_T^{4\ell}\text{-Low}$ category and (b) the BDT distribution of the $1j\text{-}p_T^{4\ell}\text{-Low}$ category, in a non-resonant ZZ^* -enriched control region, obtained by applying the full event selection except for the $m_{4\ell}$ window, i.e. $m_{4\ell} < 115$ GeV or $130 < m_{4\ell} < 170$ GeV. The error bars on the data points indicate the statistical uncertainty. The uncertainty in the prediction is shown by the dashed band.

Table 10: The expected number of SM Higgs boson events with $m_H = 125$ GeV in the mass range $115 < m_{4\ell} < 130$ GeV at an integrated luminosity of 79.8 fb^{-1} and $\sqrt{s} = 13$ TeV in each reconstructed event category. Statistical and systematic uncertainties are reported together.

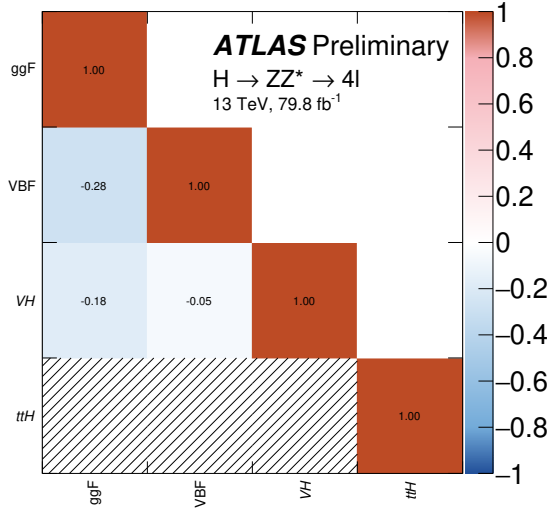
Reconstructed Category	SM Higgs boson production mode								Total	Observed
	ggF+bbH	VBF	VH	ttH + tH	ZZ*	RedBkg	ttV			
0-jet $p_T^{4\ell} < 100$ GeV	55 ± 5	0.64 ± 0.12	0.41 ± 0.04	0.00057 ± 0.00014	35.2 ± 2.5	4.1 ± 0.4	0.037 ± 0.006		95 ± 6	112
1-jet $p_T^{4\ell} < 60$ GeV	16.4 ± 2.2	1.05 ± 0.06	0.46 ± 0.04	0.0031 ± 0.0010	7.6 ± 1.1	1.31 ± 0.14	0.050 ± 0.008		26.9 ± 2.5	25
1-jet $60 \text{ GeV} < p_T^{4\ell} < 120$ GeV	9.7 ± 1.5	1.38 ± 0.15	0.49 ± 0.04	0.0040 ± 0.0012	2.25 ± 0.32	0.67 ± 0.08	0.051 ± 0.011		14.5 ± 1.5	14
1-jet $p_T^{4\ell} > 120$ GeV	2.5 ± 0.5	0.60 ± 0.07	0.222 ± 0.018	0.0035 ± 0.0007	0.31 ± 0.05	0.171 ± 0.027	0.027 ± 0.005		3.8 ± 0.5	4
2-jet VBF-enriched $p_T^{4\ell} < 200$ GeV	7.9 ± 1.6	4.1 ± 0.4	0.64 ± 0.06	0.179 ± 0.015	2.8 ± 0.7	0.74 ± 0.09	0.30 ± 0.04		16.6 ± 1.8	24
2-jet VH-enriched	5.5 ± 1.1	0.43 ± 0.04	1.20 ± 0.08	0.068 ± 0.009	1.6 ± 0.4	0.47 ± 0.06	0.125 ± 0.017		9.4 ± 1.2	11
2-jet VBF-enriched $p_T^{4\ell} > 200$ GeV	0.70 ± 0.20	0.38 ± 0.04	0.112 ± 0.013	0.050 ± 0.005	0.28 ± 0.09	0.115 ± 0.024	0.040 ± 0.007		1.68 ± 0.23	4
VH-leptonic-enriched	0.033 ± 0.004	0.0084 ± 0.0004	0.56 ± 0.04	0.100 ± 0.011	0.068 ± 0.013	0.011 ± 0.008	0.0242 ± 0.0033		0.80 ± 0.05	1
0-jet $p_T^{4\ell} > 100$ GeV	0.059 ± 0.022	0.0096 ± 0.0017	0.114 ± 0.011	0.00032 ± 0.00016	0.0082 ± 0.0027	0.147 ± 0.023	0.018 ± 0.004		0.355 ± 0.034	0
ttH-hadronic-enriched	0.10 ± 0.09	0.020 ± 0.004	0.041 ± 0.007	0.44 ± 0.04	0.035 ± 0.016	0.072 ± 0.018	0.122 ± 0.023		0.83 ± 0.11	0
ttH-leptonic-enriched	0.0007 ± 0.0007	0.00017 ± 0.00005	0.0044 ± 0.0007	0.233 ± 0.026	0.0005 ± 0.0005	0.0028 ± 0.0026	0.0117 ± 0.0021		0.253 ± 0.026	0
Total	98 ± 8	8.6 ± 0.4	4.25 ± 0.30	1.08 ± 0.10	50 ± 4	8.16 ± 0.07	0.81 ± 0.10		171 ± 8	195



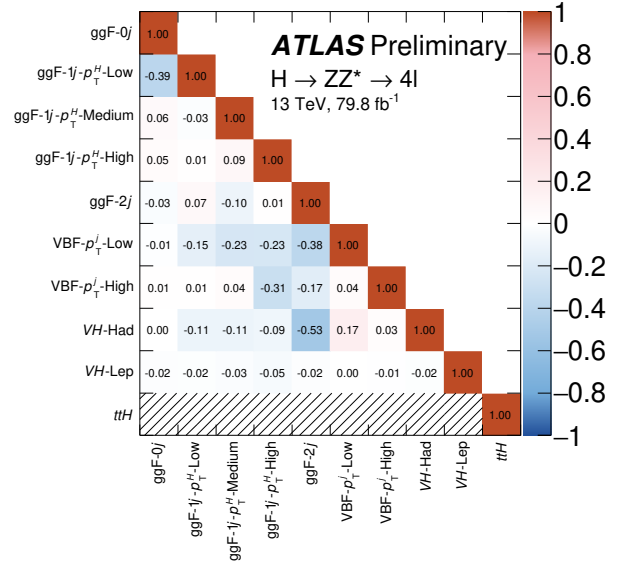
(a) Stage 0, expected



(b) Reduced Stage 1, expected

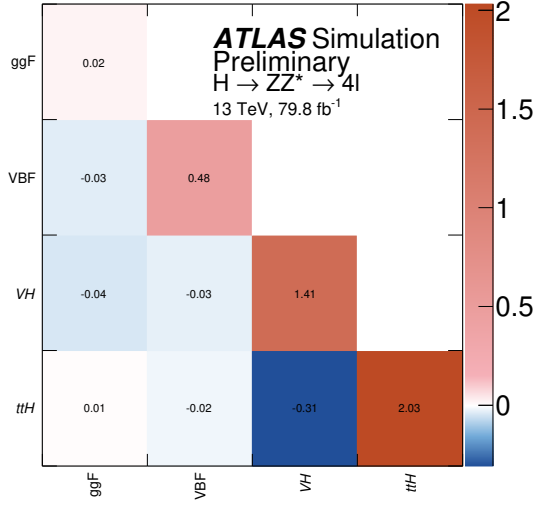


(c) Stage 0, observed

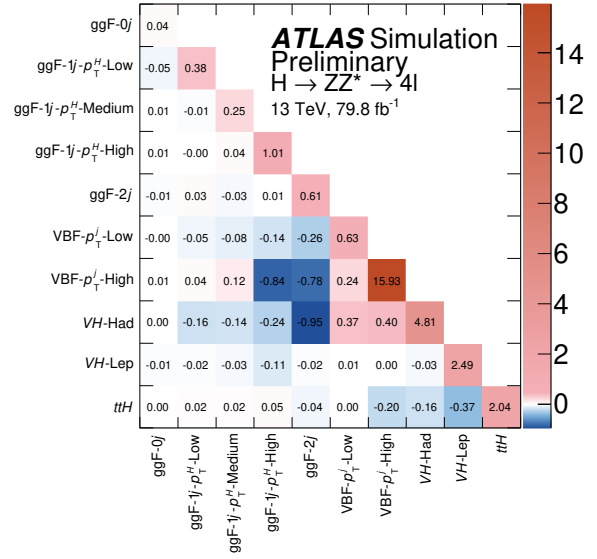


(d) Reduced Stage 1, observed

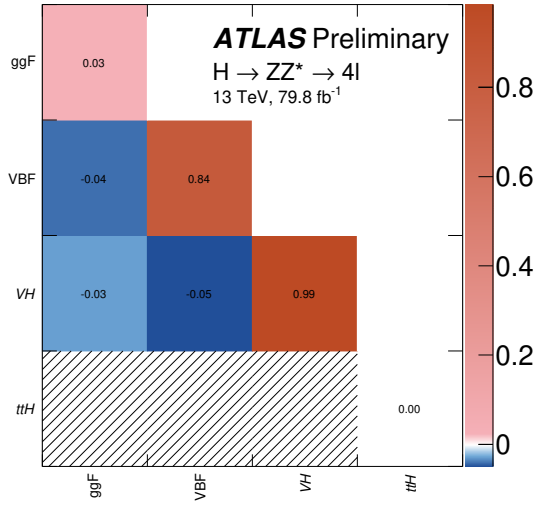
Figure 19: Correlation matrix for the different parameters of interest when fitting the Asimov dataset for (a) Stage-0 and (b) reduced Stage-1 signal strength parameters and when fitting the data for (c) Stage-0 and (d) reduced Stage-1.



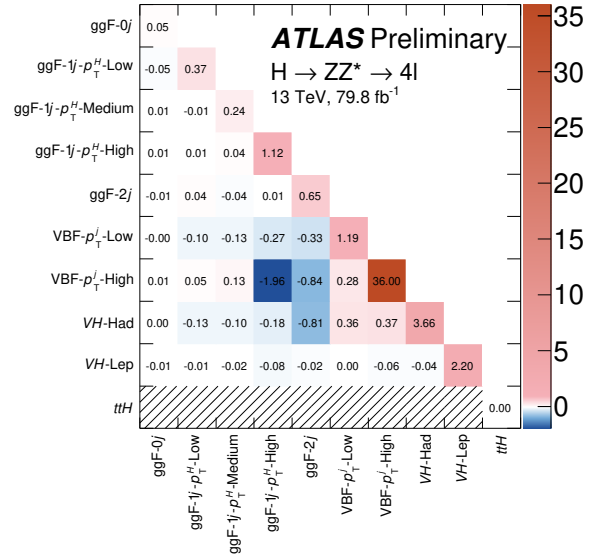
(a) Stage 0, expected



(b) Reduced Stage 1, expected



(c) Stage 0, observed



(d) Reduced Stage 1, observed

Figure 20: Covariance matrix for the different parameters of interest when fitting the Asimov dataset for (a) Stage-0 and (b) reduced Stage-1 signal strength parameters and when fitting the data for (c) Stage-0 and (d) reduced Stage-1.